Optimal Transport: Theory & Implementations

by

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A Thesis Submitted in Partial Fulfilment of
the Requirements for the Degree of

Bachelor of Science in Mathematics (Honors)

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THE UNIVERSITY OF NEW BRUNSWICK

April, 2026

Abstract

This thesis studies the Optimal Transport Problem (OTP), which seeks the most cost-efficient way to transform one probability distribution into another. It introduces the mathematical foundations required for optimal transport, including measure theory, convexity, and optimization. Two main formulations are explored: the Monge formulation, based on transport maps, and the Kantorovich formulation, which generalizes the problem by allowing mass splitting.

The existence and properties of optimal solutions are examined, along with special cases where explicit solutions can be obtained in one-dimensional and discrete settings. The dual formulation is also presented through Kantorovich duality, offering an alternative perspective on the problem. Computational methods, including linear programming and the Sinkhorn algorithm, are discussed for practical implementation. These methods are applied to a colour transfer problem, demonstrating the effectiveness of optimal transport in real-world applications.

Dedication

This thesis is dedicated to the UNB Rock & Ice Climbing Club (est. 1976).

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Chapter 1

The Optimal Transport Problem

Imagine you have a pile of sand. It has a shape and a volume. Now imagine that you have a hole in the ground. It also has a shape, which might be quite different than the sand pile's. However, this hole has the same volume as the sand. You wish to move the sand to the hole, so that the hole is completely filled with sand.

There is a cost associated with moving the sand to the hole. This could be some combination of time, equipment, money, or effort. This cost depends on where in the pile you took the sand from, and where in the hole you're putting it. There are different methods you can take to move the sand. For example, you could start by taking sand from the side of the pile nearest to the hole, then dumping it in the nearest unfilled part of the hole. Or, you could move the nearest sand to the furthest unfilled part of the hole.

There might be many approaches to take, and each one will often produce a different total cost for moving all of the sand. Obviously, you'd like to find the approach that produces the lowest total cost.

This is the basic set up for the Optimal Transport Problem (OTP). In general, we'd start with a probability distribution. We'll call this μ. For example, μ could be a uniform distribution. The important thing to know about probability distributions is that their total measure, which could also be called area or volume, equals 1. Then we have a desired probability distribution, ν. For example, ν could be the normal distribution. We wish to transform μ into ν. Note that μ is a distribution over space X, and ν is a distribution over space Y. For each $x \in \mu(X)$, there is some cost to move it to a $y \in \nu(Y)$. This can be found with a cost function, $c(x, y)$, which, for example, we can set to something like $c(x, y) = |x - y|$.

If this problem is solvable, there should be a collection of functions that map $x \in \mu(X)$ to $y \in \nu(Y)$. One of these functions should produce a minimal total cost compared to the others. We wish to find this function, as well as the total cost. To do that, we'll need to cover some more essentials first.

Chapter 2

Preliminaries

2.1 Probability & Distributions

The following definitions about basic probability spaces was taken from [4, §1.1]

Definition 2.1.1. Let Ω be the set of all possible outcomes of an experiment. Let $\mathcal{F}$ be a σ**-algebra** on Ω. That means $\mathcal{F}$ is a collection of subsets of Ω such that the following rules hold:

1. $\Omega \in \mathcal{F}$.

2. If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$.

3. If $A_1, A_2, \ldots \in \mathcal{F}$, then

$$\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}. \tag{2.1.1}$$

Definition 2.1.2. Let $\mathcal{F}$ be a σ-algebra, and μ be a function such that $\mu : \mathcal{F} \to \mathbb{R}$. μ is a **probability measure** if the following rules hold:

1. For any $A \in \mathcal{F}$, $\mu(A) \geq 0$.

2. $\mu(\Omega) = 1$ and $\mu(\emptyset) = 0$.

3. If $\{A_j\}_{j=1}^{\infty} \subset \mathcal{F}$ is a countable collection of disjoint sets, i.e., $A_{j_1} \cap A_{j_2} = \emptyset$ for $j_1 \neq j_2$, then

$$\mu\left(\bigcup_{j=1}^{\infty} A_j\right) = \sum_{j=1}^{\infty} \mu(A_j). \tag{2.1.2}$$

If $A \in \mathcal{F}$ is an event, then $\mu(A)$ is the probability of that event. $\mathcal{P}(A)$ is the set of all probability measures on A.

Definition 2.1.3. A probability measure μ on X has **cumulative distribution function** F where

$$F(x) = \int_{-\infty}^{x} d\mu = \mu\big((-\infty, x]\big) \tag{2.1.3}$$

Definition 2.1.4. Measure μ is **atomic** at $x \in X$ if $\mu(\{x\}) > 0$. μ is **non-atomic**, or assigns no point masses, if there is no such $x \in X$.

Definition 2.1.5. A **measurable space** is an ordered pair $(X, \mathcal{F})$, where X is a set, and $\mathcal{F}$ is a σ-algebra on X. An element of $\mathcal{F}$ is $\mathcal{F}$-measurable.

Definition 2.1.6. A **measure space** is a triple $(X, \mathcal{F}, \mu)$ where X is a set, $\mathcal{F}$ is a σ-algebra on X, and μ is a measure on X.

Definition 2.1.7. Let $(X, \mathcal{F})$ be a measurable space, and $E \subset X$. Then a function $f : E \to \mathbb{R}$ is a $\mathcal{F}$-**measurable function** if and only if

$$\forall \alpha \in \mathbb{R} : \{x \in E : f(x) \leq \alpha\} \in \mathcal{F} \tag{2.1.4}$$

2.2 Functions and Sequences

Definition 2.2.1. [5] Let $f : D \to \mathbb{R}$, and $\tilde{x} \in D$. f is **lower semi-continuous** at $\tilde{x}$ if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$f(\tilde{x}) - \epsilon < f(x) \quad \forall x \in B(\tilde{x}; \delta) \cap D \tag{2.2.1}$$

f is lower semi-continuous on D if this is true for every $x \in D$.

Definition 2.2.2. A function $L : \mathbb{R}^m \to \mathbb{R}^n$ is **linear** if for any $x, y \in \mathbb{R}^m$ and for any scalar a, b, we have

$$L(ax + by) = aL(x) + bL(y) \tag{2.2.2}$$

Definition 2.2.3. A function $A : \mathbb{R}^m \to \mathbb{R}^n$ is **affine** if there is a linear function $L : \mathbb{R}^m \to \mathbb{R}^n$ and a vector $b \in \mathbb{R}^n$ such that for all $x \in \mathbb{R}^m$

$$A(x) = L(x) + b \tag{2.2.3}$$

Definition 2.2.4. A function $f : \mathbb{R}^m \to \mathbb{R}$ is **smooth** if it is infinitely differentiable.

Example 2.2.5. The derivative of any constant function is 0. The constant 0 is also a constant function. This means we can infinitely take any constant function's derivative. By definition, a constant function is smooth.

2.3 Normed Vector Spaces

Remark 2.3.1. We denote the field by $\mathbb{F}$.

Definition 2.3.2. For set V, an **addition** is a function that assigns $f + g \in V$ to each pair of elements $f, g \in V$. A **scalar multiplication** is a function that assigns $\alpha f \in V$ to each $\alpha \in \mathbb{F}$ and each $f \in V$.

Definition 2.3.3. A **vector space** over $\mathbb{F}$ is a set V, with addition and scalar multiplication defined, such that the following properties hold

1. commutativity: $f + g = g + f$ for all $f, g \in V$
2. associativity: $(f + g) + h = f + (g + h)$ and $(\alpha\beta)f = \alpha(\beta f)$ for all $f, g, h \in V$ and for all $\alpha, \beta \in \mathbb{F}$.
3. additive identity: There exists an $0 \in V$ such that $f + 0 = f$ for all $f \in V$.
4. additive inverse: For every $f \in V$, there exists $g \in V$ such that $f + g = 0$.
5. multiplicative identity: There exists a $1 \in V$ such that $1f = f$ for all $f \in V$.
6. distributive properties: $\alpha(f + g) = \alpha f + \alpha g$ and $(\alpha + \beta)f = \alpha f + \beta f$ for all $f, g \in V$ and for all $\alpha, \beta \in \mathbb{F}$.

Definition 2.3.4. A **norm** on vector space V over $\mathbb{F}$ is a function $\|\cdot\| : V \to [0, \infty)$ such that

1. positive definite: $\|f\| = 0$ if and only if $f = 0$.
2. homogeneity: $\|\alpha f\| = |\alpha|\|f\|$ for all $\alpha \in \mathbb{F}$ and for all $f \in V$.
3. triangle inequality: $\|f + g\| \leq \|f\| + \|g\|$ for all $f, g \in V$.

Definition 2.3.5. A **normed vector space** is a pair $(V, \|\cdot\|)$, where V is a vector space and $\|x\|$ is a norm on V.

2.4 Lebesgue Theory

The following section takes definitions from [2].

Definition 2.4.1. The **length**, $\ell(I)$, of open interval I is defined by

$$I = \begin{cases} b - a, & if \quad I = (a,b) \text{ for some a < b, where } a, b \in \mathbb{R} \\ 0, & if \quad I = \emptyset \\ \infty, & if \quad I = (-\infty, a), (a, \infty), \text{ or } (-\infty, \infty) \end{cases} \tag{2.4.1}$$

Definition 2.4.2. The **outer measure**, $|X|$, of set X is defined by

$$|X| = \inf\left\{\sum_{k=1}^{\infty} \ell(I_k) : I_1, ..., I_k \text{ are open intervals such that } X \subseteq \bigcup_{k=1}^{\infty} I_k\right\} \tag{2.4.2}$$

Definition 2.4.3. The smallest σ-algebra containing all open subsets of set X is the collection of Borel subsets of X. An element of this set is called a **Borel subset**.

Definition 2.4.4. The **Lebesgue measure**, denoted $\mathcal{L}$, is a measure on set X that assigns the outer measure to each Borel subset of X.

Remark 2.4.5. Note that by these definitions, for the one-dimensional interval (a, b), where $a, b \in \mathbb{R}$, $\mathcal{L}((a, b)) = b - a$.

Definition 2.4.6. Let $(X, \mathcal{F}, \mu)$ be a measure space, $0 < p < \infty$, and $f : X \to \mathbb{R}$ is $\mathcal{F}$-measurable. The **p-norm** of f, denoted $\|f\|_p$, is found by

$$\|f\|_p = \left(\int |f|^p d\mu\right)^{1/p} \tag{2.4.3}$$

Definition 2.4.7. Let $(X, \mathcal{F}, \mu)$ be a measure space, $0 < p \leq \infty$. The **Lebesgue space**, denoted $L^p(\mu)$, is the set of $\mathcal{F}$-measurable functions $f : X \to F$ such that $\|f\|_p < \infty$.

2.5 Other Spaces and Sets

Definition 2.5.1. [2, §6A] A **metric** on a nonempty set V is a function $d : V \times V \to [0, \infty)$ such that the following properties hold

1. $d(f, f) = 0$ for all $f \in V$.

2. if $f, g \in V$, and $d(f, g) = 0$, then $f = g$.

3. $d(f, g) = d(g, f)$ for all $f, g \in V$.

4. $d(f, h) \leq d(f, g) + d(g, h)$ for all $f, g, h \in V$.

A **metric space** is a pair (V, d) where V is a set and d is a metric on V.

Definition 2.5.2. Let $M = (V, d)$ be a metric space on set V, and $\{x_n\}$ be a sequence on V. $\{x_n\}$ is a **cauchy sequence** if and only if

$$\forall \epsilon > 0 : \exists N \in \mathbb{N} : \forall n, m \in \mathbb{N} : m, n \geq N : d(x_n, x_m) < \epsilon \tag{2.5.1}$$

Definition 2.5.3. A metric space is **complete** if and only if every Cauchy sequence is convergent.

Definition 2.5.4. A function f is **injective** if and only if

$$\forall x_1, x_2 \in Dom(f) : f(x_1) = f(x_2) \implies x_1 = x_2 \tag{2.5.2}$$

which means the function's output uniquely determines it's input.

Definition 2.5.5. A set S is **countable** if and only if there exists an injection

$$f : S \to \mathbb{N} \tag{2.5.3}$$

Definition 2.5.6. Let $M = (V, d)$ be a metric space, $A \subset V$, and $\alpha \in V$. α is a **limit point** of A if and only if there is a sequence $\{\alpha_n\}$ in $A \setminus \{\alpha\}$ such that $\{\alpha_n\}$ converges to α.

Definition 2.5.7. Let $M = (V, d)$ be a metric space, and $A \subset V$. A is **dense** in M if and only if every point of V is a limit point of A.

Definition 2.5.8. Let $(X, \mathcal{F}, \mu)$ be a measure space. The set $N \in \mathcal{F}$ is a μ**-null set** if and only if $\mu(N) = 0$.

Definition 2.5.9. Let $(X, \mathcal{F}, \mu)$ be a measure space. Property $P(x)$ of X holds μ**-almost everywhere** if the set $\{x \in X : P(x) \text{ is false}\}$ is contained in a null set.

Definition 2.5.10. [13, p. ii] The **support** of probability measure μ is the smallest closed set X such that $\mu(X) = 1$. The support of μ is referred to as $\text{supp}(\mu)$.

Definition 2.5.11. Let S be a set. A **cover** of S is a set of sets $\mathcal{C}$ such that $S \subset \bigcup C$. If all the elements of $\mathcal{C}$ are open, then $\mathcal{C}$ is an **open cover**. A **subcover** of $\mathcal{C}$ for S is a set $\mathcal{S} \subset \mathcal{C}$ such that $\mathcal{S}$ is also a cover for S. If any cover is has finitely many elements, it is a **finite cover**.

Definition 2.5.12. A metric space $M = (V, d)$ is **compact** if and only if every open cover for V has a finite subcover.

Theorem 2.5.13. *[11][Heine-Borel] The subspace of* $\mathbb{R}^n$ *is compact if and only if it is closed and bounded.*

2.6 Convexity and Optimization

Definition 2.6.1. [9] Let V be a vector space over $\mathbb{F}$, and $C \subseteq V$. C is a **convex set** if and only if

$$tx + (1 - t)y \in C \tag{2.6.1}$$

for each $x, y \in C$ and $t \in [0, 1]$.

Definition 2.6.2. [13, p. ii] A function $f : \mathbb{R}^d \to \mathbb{R}$ is a **convex function** if for all $x, y \in \mathbb{R}^d$ and all $\theta \in [0, 1]$, we have

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y) \tag{2.6.2}$$

Definition 2.6.3. [7, §2.1.3]. A function f is m-**strongly convex** if $f(\theta P + (1 - \theta)Q) \leq \theta f(P) + (1 - \theta)f(Q) - \frac{m}{2}\theta(1 - \theta)\|P - Q\|_2^2$ for all P, Q, some $m \geq 0$, and $\theta \in [0, 1]$. Note that all strongly convex functions are trivially convex functions.

Definition 2.6.4. [12] A function f is **strictly convex** if $f(\theta P + (1 - \theta)Q) < \theta f(P) + (1 - \theta)f(Q)$ for all P, Q, and $\theta \in (0, 1)$. Note that all strongly convex functions are strictly convex functions.

Remark 2.6.5. [7, §2.1.1] claims that if $-f$ is convex, then f is concave. We apply the same rule to the definitions of strongly and strictly convex, as well as theorems relating to convexity and optimization.

Theorem 2.6.6. *[10] Let $K \subset X$ be the compact subset of a normed vector space. Let f be continuous on K. Then f attains both a minimum and a maximum on K.*

Definition 2.6.7. Let f be defined on open interval (a, d). Let $x \in (a, b)$. f has a **local minimum** at x_0 if and only if

$$\exists (b, c) \subset (a, d) : \forall x \in (b, c) : f(x) \geq f(x_0) \tag{2.6.3}$$

Theorem 2.6.8. *All local minimizers of a convex function are global minimizers [7, §2.1.1]. The global minimizer of a strongly convex function is unique [7, §2.1.3].*

Proof of theorem 2.6.8. Let f be convex. Suppose x^* is a local minimizer, but not a global one. That means there exists some y such that $f(y) < f(x^*)$. Let $z_\theta =$

$\theta y+(1-\theta)x^*$ where $\theta \in [0,1]$. By convexity and hypothesis, we have $f(z_\theta) \leq \theta f(y)+(1-\theta)f(x^*) < \theta f(x^*)+(1-\theta)f(x^*) = f(x^*)$. As $\theta \to 0$, we have $z_\theta \to x^*$, while still maintaining $f(z_\theta) < f(x^*)$. But this violates the definition of local minimizer, so x^* must be a global minimizer. This means all local minimizers of a convex function are global minimizers.

Let f be strongly convex. Suppose x_1 and x_2 are different global minimizers, with minimum value f_{min}. By definition of strongly convex, we see $f(\theta x_1 + (1-\theta)x_2) \leq \theta f(x_1)+(1-\theta)f(x_2) - \frac{m}{2}\theta(1-\theta)\|x_1-x_2\|^2 = f_{min} - \frac{m}{2}\theta(1-\theta)\|x_1-x_2\|^2$. Note that since $x_1 \neq x_2$, $\|x_1 - x_2\| > 0$, so we can rearrange the inequality to find $f(\theta x_1 + (1-\theta)x_2) < f_{min}$. This contradicts our assumption that f_{min} was the smallest possible value f could take. Therefore, the global minimizer to a strongly convex function must be unique $\square$

2.7 Bregman Theory

The following definitions and theorem come from [3, §1]

Definition 2.7.1. Let $f : \mathbb{R}^n \to \mathbb{R}$ be a differentiable and strictly convex function. The **Bregman divergence** of f between x and y is $D_f(x\|y) = f(x) - f(y) - \nabla f(y)^T(x-y)$, where $\nabla f(y)$ is a vector of partial derivatives where entry $\nabla f(y)_i = \frac{\partial f}{\partial x_i}(y)$.

Definition 2.7.2. Let $C \subset \mathbb{R}^n$ be closed and convex set. The **Bregman projection** of point y onto C is $\Pi_C^f(y) = \arg\min_{x\in C} D_f(x\|y)$. Because f is strictly convex, the projection is unique. If this projection can be found algebraically, we call that the closed form (KL) Bregman projection.

Theorem 2.7.3 (Alternating Bregman Divergence)**.** *Let $A, B \subset \mathbb{R}^n$ be closed and convex sets with a non-empty intersection. Let $f : \mathbb{R}^n \to \mathbb{R}$ be a differentiable and* *strictly convex function. Start from any x_0 in the domain. Define the following sequences $x_{2k+1} = \Pi_A^f(x_{2k})$, where Π_A^f is the Bregman projection as defined above, and $x_{2k+2} = \Pi_B^f(x_{2k+1})$. This is repeatedly projecting A and B onto each other.*

x_k will converge to some $x^* \in A \cap B$, which is the unique minimizer $x^* = \arg\min_{x \in A \cap B} D_f(x \| x_0)$. This is the point in the intersection that minimizes the Bregman divergence to x_0.

2.8 Lagrangian Theory

This section comes from [6]

Theorem 2.8.1. *Let $U \subset \mathbb{R}^n$ be open, and let $f : U \to \mathbb{R}$ be a smooth* ***objective function****. Let $g_j : U \to \mathbb{R}$, for $1 \leq j \leq r$, be smooth* ***constraint functions*** *whose gradients are linearly independent at a point of interest.*

Suppose we want to optimize $f(x)$ for $x \in U$ subject to the constraints

$$g_j(x) = c_j, \quad 1 \leq j \leq r.$$

The constants c_j are called the ***constraint values****.*

Let $\mathbf{a} = (a_1, \ldots, a_n)$ be a local extremum of f subject to these constraints. Then there exist scalars $\lambda = (\lambda_1, \ldots, \lambda_r)$, called ***Lagrange multipliers****, such that*

$$\nabla f(\mathbf{a}) = \sum_{j=1}^{r} \lambda_j \nabla g_j(\mathbf{a}). \tag{2.8.1}$$

Equivalently, define the ***Lagrangian*** *function*

$$\mathcal{L}(x_1, \ldots, x_n, \lambda_1, \ldots, \lambda_r) = f(x_1, \ldots, x_n) - \sum_{j=1}^{r} \lambda_j \big(g_j(x_1, \ldots, x_n) - c_j\big). \tag{2.8.2}$$

Any constrained extremum occurs at a solution of the system

$$\frac{\partial \mathcal{L}}{\partial x_i}(x_1, \ldots, x_n, \lambda_1, \ldots, \lambda_r) = 0, \quad 1 \leq i \leq n, \tag{2.8.3}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_j}(x_1, \ldots, x_n, \lambda_1, \ldots, \lambda_r) = 0, \quad 1 \leq j \leq r. \tag{2.8.4}$$

The equations (2.8.3) *are equivalent to the gradient condition* (2.8.1)*, while* (2.8.4) *recover the original constraints.*

Chapter 3

Formulations of the OTP

Let μ and ν be probability measures on spaces X and Y, respectively. Optimal transport builds a framework to compare these two measures by some cost between them. There are two ways to formulate the optimal transport problem: the Monge and Kantorovich formulations.

3.1 The Monge Formulation

The following definitions and results come from [13, §1.1].

Definition 3.1.1. Let $T\colon X \to Y$ be a function. The map T is called a **transport map** if for all measurable sets B

$$\nu(B) = \mu\big(T^{-1}(B)\big) \tag{3.1.1}$$

If this is held, we can say $\nu = T_{\#}\mu$. ν is then called the **pushforward measure**.

Example 3.1.2. Let X have distribution $\mu \sim Uniform(0,1)$, and Y has distribution ν. Let $T : X \to Y$ by $T(x) = 2x$ be a transport map from μ to ν. The pushforward of μ is $\nu = T_{\#}\mu$ by definition.

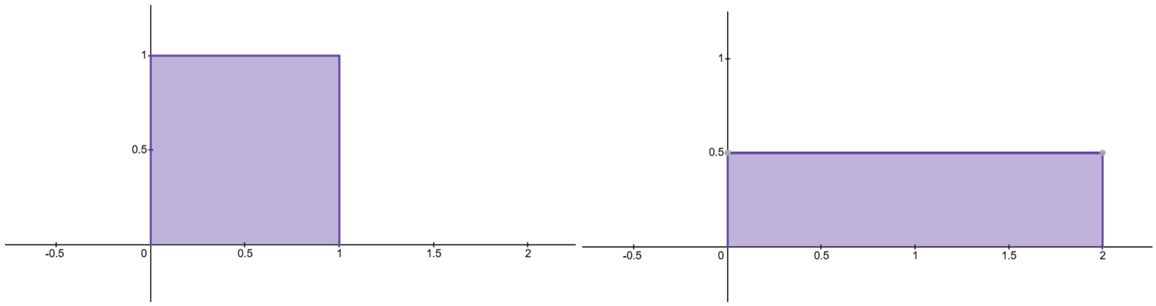

Figure 3.1: $\mu \sim$ uniform(0,1) and $\nu \sim$ uniform(0,2)

For some interval $B = [a, b]$, we have $T^{-1}(B) = \{x : 2x \in [a, b]\} = [\frac{a}{2}, \frac{b}{2}]$. So $\mu([\frac{a}{2}, \frac{b}{2}]) = \frac{b-a}{2}$, which means $\nu([a, b]) = \frac{b-a}{2}$. We can confirm that $\nu = T_{\#}\mu = uniform(0, 2)$.

Theorem 3.1.3 (Composition of Maps)**.** *Let $\mu \in \mathcal{P}(X), T : X \to Y$ and $S : Y \to Z$. Then,*

$$(S \circ T)_{\#}\mu = S_{\#}(T_{\#}\mu) \tag{3.1.2}$$

Theorem 3.1.4 (Change of Variables)**.** *[see [13, §1.1]] Let $\mu, X, T, Y, S,$ and Z be defined as in 3.1.3. If $f \in L^1(Y)$, then we also have*

$$\int_Y f(y) d(T_{\#}\mu)(y) = \int_X f(T(x)) d\mu(x) \tag{3.1.3}$$

Definition 3.1.5. Let c be a function such that $c : X \times Y \to [0, +\infty]$. c is **the cost function**, where $c(x, y)$ measures the cost of moving one unit of mass from $x \in X$ to $y \in Y$.

Definition 3.1.6 (Monge's Optimal Transport Problem)**.** Let $\mu \in \mathcal{P}(X), \nu \in \mathcal{P}(Y)$ and $T : X \to Y$. We wish to minimize the total cost of moving every $x \in X$ to a $y \in Y$. **Monge's Optimal Transport Problem** defines the minimal cost as

$$\min_{\nu = T_{\#\mu}} \mathbb{M}(T) = \int_X c(x, T(x))\, d\mu(x). \tag{3.1.4}$$

3.2 The Kantorovich Formulation

Note that in the Monge formulation, each $x \in X$ is mapped to only a single $y \in Y$. Consider the following problem. Let μ be the discrete-uniform distribution with all its mass at one point, and ν be the discrete-uniform distribution with its mass evenly distributed at two points. No matter which y we send the mass at x to, the problem is unsolved; there is no valid transport map.

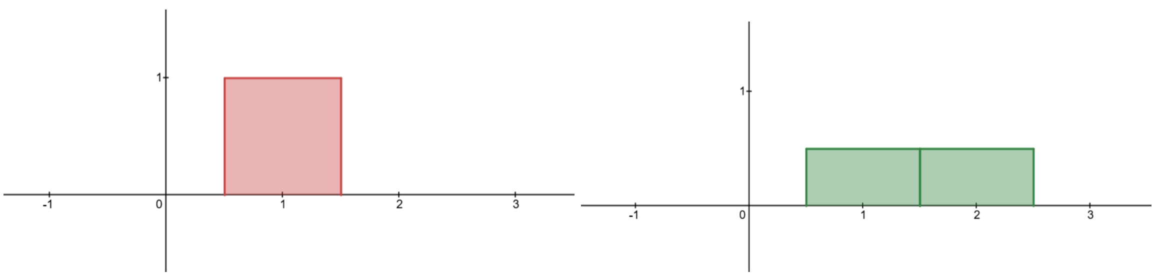

Figure 3.2: $\mu \sim$ uniform(1) and $\nu \sim$ uniform(1, 2)

The Kantorovich formulation is a relaxation of the Monge formulation that addresses this problem. By allowing for mass-splitting, we can always find an optimal solution. Instead of a transport map that tells you which $y \in Y$ each $x \in X$ is sent to, or a non-function which returns multiple $y \in Y$ for each $x \in X$, we use a function that tells you how much of the mass at each $x \in X$ is sent to each $y \in Y$.

The following definitions and results come from [13, §1.2].

Definition 3.2.1. Let there be measure π defined on $X \times Y$ such that for all measurable sets $A \subseteq X$ and $B \subseteq Y$,

$$\pi(A \times Y) = \mu(A) \quad \text{and} \quad \pi(X \times B) = \nu(B) \tag{3.2.1}$$

Then π is a **transport plan**.The set of all transport plans between μ and ν is denoted as $\Pi(\mu, \nu)$. This set is never empty, as there exists a trivial transport plan that splits and transports each x to each y proportional to $\nu(y)$.

Definition 3.2.2 (Kantorovich's Optimal Transport Problem)**.** Let $\mu \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Y)$. **Kantorovich's Optimal Transport Problem** defines the minimal cost of transportation as

$$\min_{\pi \in \Pi(\mu,\nu)} \mathbb{K}(\pi) = \int_{X \times Y} c(x,y)\, d\pi(x,y) \tag{3.2.2}$$

Theorem 3.2.3. *Every transport map is a transport plan. This means if an optimal Monge solution exists, the optimal Kantorovich solution will be at most the optimal Monge solution, i.e.*

$$\min_{\pi \in \Pi(\mu,\nu)} \mathbb{K}(\pi) \leq \min_{T : T_{\#}\mu = \nu} \mathbb{M}(T) \tag{3.2.3}$$

3.3 The Existence of Kantorovich OTP Solutions

The following two definitions come from [13, p. i].

Definition 3.3.1. The sequence of probability measures π_n on $X \times Y$ **weakly converges** if for all continuous and bounded functions $f : (X \times Y) \to \mathbb{R}$

$$\int_{X \times Y} f(x,y) d\pi_n(x,y) \to \int_{X \times Y} f(x,y) d\pi(x,y) \tag{3.3.1}$$

Definition 3.3.2 (see [13, p. i])**.** A **Polish space** is a complete metric space with a countable dense subset.

The following theorems, definitions, and proof comes from [13, §1.3].

Theorem 3.3.3. *If X and Y are Polish spaces, and c is lower semi-continuous, then there exists a $\pi^* \in \Pi(\mu, \nu)$ that minimizes $\mathbb{K}$.*

Recall that $\Pi(\mu, \nu)$ is non-empty. Before beginning the proof, note these definitions and theorems.

Definition 3.3.4. Let E be a Polish space, and $\mathcal{F} \subset \mathcal{P}(E)$. $\mathcal{F}$ is **tight** if for all $\epsilon > 0$, there exists a compact set $K_\epsilon \subset E$ such that $\mu(K_\epsilon) \geq 1 - \epsilon$ for all $\mu \in \mathcal{F}$.

Theorem 3.3.5 (Prokhorov's)**.** *Let E and $\mathcal{F}$ be defined as above. If $\mathcal{F} \subset \mathcal{P}(E)$ is tight, then $\bar{\mathcal{F}}$ is weakly compact.*

Theorem 3.3.6 (Portmanteau)**.** *Let S be a metric space, and $\mu_n \to \mu$ be a sequence of probability measures on S. For every lower semi-continuous function $f : S \to (-\infty, \infty]$ that is bounded from below, we have*

$$\int_S f d\mu \leq \lim_{n \to \infty} \inf \int_S f d\mu_n \tag{3.3.2}$$

Proof of 3.3.3. Fix $\delta > 0$. Then choose compact sets $K \subset X, L \subset Y$ such that $\mu(X \backslash K) \leq \delta$ and $\nu(Y \backslash L) \leq \delta$. If $(x, y) \notin K \times L$, then either $x \notin K$ or $y \notin L$. So

$$(X \times Y) \backslash (K \times L) \subset (X \times (Y \backslash L)) \cup ((X \backslash K) \times Y)$$

So for any $\pi \in \Pi$, we have

$$\pi((X \times Y) \backslash (K \times L)) \leq \pi(X \times (Y \backslash L)) + \pi((X \backslash K) \times Y)$$

For any measurable sets A and B, $\pi(X \times B) = \nu(B)$ and $\pi(Y \times A) = \mu(A)$. Let $A = X \backslash K$ and $B = Y \backslash L$. So,

$$\pi((X \times Y)\backslash(K \times L)) \leq \nu(Y\backslash L) + \mu(X\backslash K) \leq 2\delta$$

So for any $\delta > 0$ we have a compact set $K \times L$ so that every π assigns at least $1-2\delta$ on it. By definition, Π is tight. By Prokhorov's theorem, $\bar{\Pi}$ is weakly compact.

Let π_n be a sequence where $\pi_n \in \Pi$ for every n. Let $f(x,y) = \tilde{f}(x)$ where $\tilde{f}$ is continuous, bounded, and dependent only on x. Let $P_X(x,y) = x$. First note that from the definition of a pushback function and change of variables formula,

$$\int_{X\times Y} \tilde{f} d\pi_n(x,y) = \int_X \tilde{f}(x) d((P_X)_{\#}\pi_n)(x) = \int_X \tilde{f}(x) d\mu(x)$$

This means that for every n, $\int_{X\times Y} \tilde{f} d\pi_n(x,y) = \int_X \tilde{f} d\mu(x)$. Then, take this limit,

$$\lim_{n\to\infty} \int_{X\times Y} \tilde{f} d\pi_n(x,y) = \int_{X\times Y} \tilde{f} d\pi(x,y)$$

Note that since the left side of the above equality is equal to $\int_X \tilde{f} d\mu(x)$ for every n

$$\lim_{n\to\infty} \int_{X\times Y} \tilde{f} d\pi_n(x,y) = \int_X \tilde{f}(x) d\mu(x)$$

Combining these all together, we see

$$\int_X \tilde{f}(x) d\mu(x) = \int_{X\times Y} \tilde{f} d\pi(x,y) = \int_{X\times Y} \tilde{f} d(P_{X\#}\pi)(x)$$

Which means $P_{X\#}\pi = \mu$. Since this is true for all $\tilde{f}$ that are continuous and bounded on X, and similarly true that for all $\tilde{f} \in C_b(Y)$ we have $P_{Y\#}\pi = \nu$, this means that $\pi \in \Pi$. So Π is weakly closed, which means it is equal to its closure. So Π is weakly sequentially compact.

Now, pick π_n such that $\mathbb{K}(\pi_n) \to \inf_{\pi \in \Pi} \mathbb{K}(\pi)$. Because Π is compact, we have a subsequence such that $\pi_n \to \pi^t \in \Pi$, where π^t is a candidate minimizer. By the Portmanteau theorem,

$$\min_{\pi \in \Pi} \mathbb{K}(\pi) = \lim_{n \to \infty} \int c d\pi_n \geq \int c d\pi^t = \mathbb{K}(\pi^t) \tag{3.3.3}$$

But we always have $\mathbb{K}(\pi^t) \geq \min_{\pi \in \Pi} \mathbb{K}(\pi)$, so with these inequalities we can conclude that $\mathbb{K}(\pi^t) = \min_{\pi \in \Pi} \mathbb{K}(\pi)$. Therefore, π^t is a minimizer. $\square$

Chapter 4

Special Cases

Most OTPs require complex algorithms with additional theory to solve. Some of these methods will be covered later. In certain instances, OTPs may require only a few simple steps to be solved, irrespective of the complexity of the input or target distributions. Two of these types of special cases will be outlined in this chapter.

4.1 One-Dimensional Optimal Transport

The following definitions and results come from [13, §2.1]. This section covers how optimal transport between two continuous and one-dimensional distributions can be made simple.

Definition 4.1.1. Let μ have cumulative distribution function F. The **generalized inverse** of F on $[0, 1]$ is defined as

$$F^{-1}(t) = \inf\{x \in \mathbb{R} : F(x) > t\} \tag{4.1.1}$$

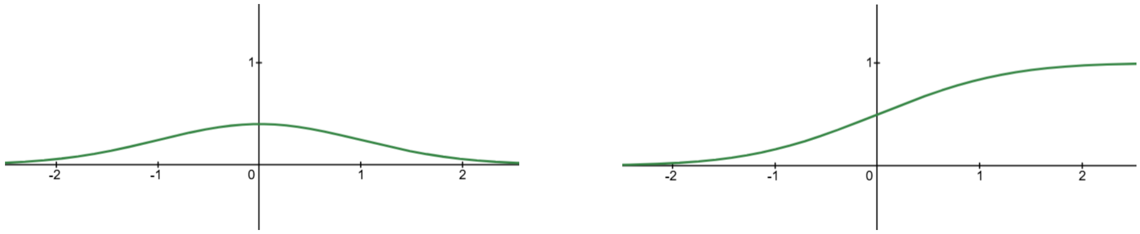

Figure 4.1: Normal distribution and its corresponding cumulative distribution

Figure 4.2: Discrete distribution and its corresponding cumulative distribution

Figure 4.3: Inverse CDF of normal distribution from figure 4.1 (green) and inverse CDF of discrete distribution from figure 4.2 (red)
.

Let ν similarly have G and G^{-1}. Since cumulative distribution functions (CDFs) are non-decreasing, it is possible for their inverses to be discontinuous.

Theorem 4.1.2. *Let the cost function, $c : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, be convex, be continuous, and be representative of the difference of two distributions. Let H be the cumulative distribution function (CDF) defined by $H(x, y) = \min\{F(X), G(X)\}$. π^t is the measure on $\mathbb{R}^2$ with CDF H. Let $\pi^t \in \Pi(\mu, \nu)$ be the optimal transport plan. The optimal cost can be found with*

$$\min_{\pi \in \Pi(\mu,\nu)} \mathbb{K}(\pi) = \int_0^1 c(F^{-1}(t), G^{-1}(t))dt \tag{4.1.2}$$

And if $c(x, y) = |x - y|$, then

$$\min_{\pi \in \Pi(\mu,\nu)} \mathbb{K}(\pi) = \int_{\mathbb{R}} |F(x) - G(x)|\, dx \tag{4.1.3}$$

If the source distribution μ has no point masses, then the optimal Kantorovich and Monge solutions are the same. Further, let $T^t = G^{-1}(F)$. Then T^t is the solution to the Monge OTP.

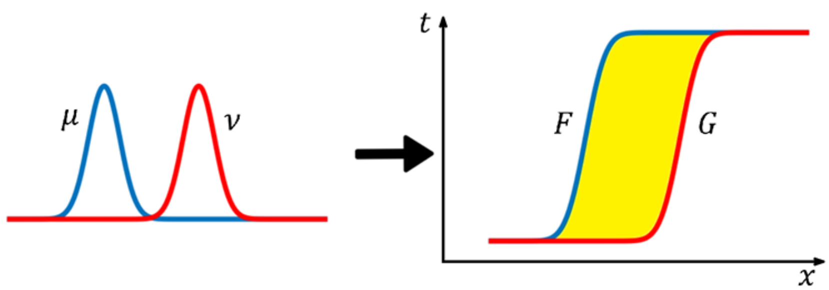

Figure 4.4: Overlaid distributions and their overlaid cumulative distributions

4.1.1 example

Example 4.1.3. Let $\mu = \text{Uniform}(0, 1)$ and $\nu = \text{Uniform}(1, 2)$, with cost function $c(x, y) = |x - y|$.

The cumulative distribution functions are

$$F(x) = \begin{cases} 0, & x < 0, \\ x, & 0 \le x \le 1, \\ 1, & x > 1, \end{cases} \qquad G(x) = \begin{cases} 0, & x < 1, \\ x - 1, & 1 \le x \le 2, \\ 1, & x > 2. \end{cases}$$

The generalized inverses are $F^{-1}(t) = t$ and $G^{-1}(t) = t + 1$ for $0 \leq t \leq 1$. By Theorem 4.1.2

$$\min_{\pi \in \Pi(\mu,\nu)} \mathbb{K}(\pi) = \int_0^1 |F^{-1}(t) - G^{-1}(t)|\, dt = \int_0^1 |t - (t+1)|\, dt = \int_0^1 1\, dt = 1.$$

Since μ has no point masses, the optimal Monge map is $T(x) = G^{-1}(F(x))$. For $x \in [0, 1]$, we have $F(x) = x$, so $T(x) = G^{-1}(x) = x + 1$. We also compute

$$\int_{\mathbb{R}} |F(x) - G(x)|\, dx = \int_0^1 x\, dx + \int_1^2 (2 - x)\, dx = \frac{1}{2} + \frac{1}{2} = 1.$$

The optimal transport map is $T(x) = x + 1$ and the optimal transport cost is

$$\min_{\pi \in \Pi(\mu,\nu)} \mathbb{K}(\pi) = 1.$$

4.1.2 Proof of One-Dimensional Solution Formula

Proof of theorem 4.1.2. Assume conditions of 4.1.2. Recall the composition of transport maps theorem, $T_{\#}\mu = G^{-1}_{\#}(F_{\#}\mu)$. By definition of pushforward, $G^{-1}_{\#}\mathcal{L}_{[0,1]}((-\infty, y]) = \mathcal{L}_{[0,1]}(t : G^{-1}(t) \leq t)$, where $\mathcal{L}_{[0,1]}(A)$ is the Lebesgue measure of set A restricted to $[0, 1]$. Then,

$$\mathcal{L}_{[0,1]}(t : G^{-1}(t) \leq y) = \mathcal{L}_{[0,1]}(t : G(y) \geq t) = \mathcal{L}_{[0,1]}([0, G(y)]) = G(y) = \nu((-\infty, y])$$

So, $G^{-1}_{\#}\mathcal{L}_{[0,1]} = \nu$.

μ has no point masses, so F is continuous. Because of that $F^{-1}([0,t])$ is closed, i.e. $F^{-1}([0,t]) = \{x : F(x) \in [0,t]\} = \{x : F(x) \leq t\}$, and $F(x_t) = t$. Because F is non-decreasing, $\mu(\{x : F(x) \leq t\}) = \mu(\{x : x \leq x_t\} = \mu((-\infty, x_t]) = F(x_t)$. Now, we can calculate

$$F_{\#}\mu([0,t]) = \mu(F^{-1}([0,t])) = \mu(\{x : F(x) \leq t\} = \mu((-\infty, x_t]) = F(x_t) = t.$$

So, $F_{\#}\mu([0,t]) = t$. Note that $\mathcal{L}_{[0,1]} = t$. Since $F_{\#}\mu([0,t])$ and $\mathcal{L}_{[0,1]} = t$ agree on the set $[0,1]$, we can say that $F_{\#}\mu = \mathcal{L}_{[0,1]}$.

Returning to our earlier statement, $T_{\#}\mu = G^{-1}_{\#}(F_{\#}\mu) = G^{-1}_{\#}(\mathcal{L}_{[0,1]}) = \nu$. By definition, this means that T^t is a valid solution to the Monge formulation. We also have from equation 4.1.2

$$\min_{\pi \in \Pi} \mathbb{K}(\pi) = \int_0^1 c(F^{-1}(t), G^{-1}(t))dt$$

By change of variables, and our earlier definition that $T^t = G^{-1}(F)$,

$$\int_0^1 c(F^{-1}(t), G^{-1}(t))dt = \int_{\mathbb{R}} c(x, G^{-1}(F(x)))d\mu(x) = \int_{\mathbb{R}} c(x, T^t)d\mu(x) = \mathbb{M}(T^t)$$

Where $\mathbb{M}(T^t)$ is the Monge problem evaluated at T^t. This means $\min \mathbb{K}(\pi) = \mathbb{M}(T^t)$. Intuitively, $\mathbb{M}(T^t) \geq \min \mathbb{M}(T)$ for any $T : T_{\#}\mu = \nu$. We also have $\min \mathbb{M}(T) \geq \min \mathbb{K}(\pi)$. Putting all these together shows that $\min \mathbb{K}(\pi) = \min \mathbb{M}(T) = \mathbb{M}(T^t)$. $\square$

4.1.3 One-Dimensional Example

We now have sufficient knowledge to solve the problem that was introduced earlier. Let our starting distribution, μ, be the uniform distribution over $[0, 10]$. Let our target distribution, ν be the normal distribution with mean 5 and standard deviation 2.5. Our cost function will be $c(x, y) = |x - y|$.

First, note that these distributions are both in $\mathbb{R}$, which means that this problem is 1-dimensional. We can use some of the special cases discussed earlier to solve. First, let us identify cumulative distributional functions $F(x)$ and $G(y)$ for μ and ν respectively. Recall that

$$F(x) = \mu\big((-\infty, x]\big) \text{ and } G(y) = \nu\big((-\infty, y]\big)$$

Which means

$$F(x) = \begin{cases} 0, & x < 0 \\ x/10, & 0 \leq x \leq 10 \\ 1, & x > 10 \end{cases}$$

and

$$G(y) = \Phi(\frac{y-5}{2.5})$$

Where Φ is the CDF function of the normal distribution. Since our cost function, $c(x, y) = |x - y|$, is convex and continuous, our optimal transport plan $\pi^t \in \Pi(\mu, \nu)$ has CDF $H(x, y) = \min\{F(x), G(y)\}$. Note that this transport plan doesn't depend on c directly. As long as our cost is convex and continuous, the optimal transport plan will have the same CDF. By our earlier definition of CDF, $H(x, y) = \pi^t((-\infty, x) \times (-\infty, y))$. This means we can find out how much mass is transported from $(a, b] \in X$ to $(c, d] \in Y$ by calculating

$$\pi^t((a,b]\times(c,d]) = H(b,d) - H(a,d) - H(b,c) + H(a,c)$$

$$= \min\{F(b),G(d)\} - \min\{F(a),G(d)\} - \min\{F(b),G(c)\} + \min\{F(a),G(c)\}$$

This is a solution to the Kantorovich OTP. In many optimal transport problems, there are no Monge solutions. However, since F is continuous, μ assigns no point masses, (i.e. $\mu(\{x\}) = 0 \forall x \in X$). This means that the Monge solution exists, and the total cost it produces is equal to the total cost of the Kantorovich solution. The optimal transport map, T^t, is found with $T^t = G^{-1} \circ F$. Note that

$$G^{-1}(t) = inf\{x \in \mathbb{R} : G(y) > t\}$$

and that for $x \in [0, 10]$,

$$T^t(x) = 5 + 2.5\Phi^{-1}(x/10)$$

Since our cost function is $c(x, y) = |x - y|$, we can find the total minimal cost with

$$\int_{\mathbb{R}} |F(x) - G(x)|dx$$

$$= \int_{-\infty}^{0} |G(x)|dx + \int_{0}^{10} |F(x) - G(x)|dx + \int_{10}^{\infty} |1 - G(x)|dx \approx 0.6$$

If our cost wasn't $|c(x, y) = |x - y|$, but some other function that is convex and continuous, like $c(x, y) = |x - y|^2$, then we could still find the total cost. The new value would be

$$\int_0^1 (F^{-1}(t) - G^{-1}(t))^2 dt = \int_0^1 (10t - (5 + 2.5\Phi^{-1}(t)))^2 dt \approx 14$$

4.2 Discrete Optimal Transport

The following definitions and results come from [13, §2.2]. This section covers how optimal transport between two evenly weighted and sized discrete distributions can be made simple.

Discrete optimal transport problems often do not admit Monge solutions because the Monge formulation requires a transport map, which does not allow mass splitting. Discrete source and target atomic masses rarely align exactly, so satisfying the marginal constraints typically requires splitting mass. For this reason, we mostly focus on Kantorovich solutions for discrete OTPs.

Definition 4.2.1. Let B be a compact and convex set in $\mathbb{R}^n$. If point $b \in B$ cannot be written as a nontrivial convex combination of points in B, then b is an **extreme point** of B. The set of all these points in B is denoted $\mathcal{E}$(B). This set can be, but is not necessarily, finite.

Theorem 4.2.2 (Minkowski–Caratheodory)**.** *For every $\pi^t \in B$, there exists a probability measure η_{π^t} supported on $\mathcal{E}(B)$ such that for every affine function $f\colon B \to \mathbb{R}$,*

$$f(\pi^t) = \int_{\mathcal{E}(B)} f(\pi)\, d\eta_{\pi^t}(\pi). \tag{4.2.1}$$

In the discrete case, this becomes

$$f(\pi^t) = \sum_{i=1}^{k} \lambda_i f(\pi^i) \tag{4.2.2}$$

Where $\lambda_i = \eta_{\pi_i}(\pi^i)$ is the mass assigned to π^i.

Theorem 4.2.3 (Birkoff's Theorem)**.** *If B is the set of $n \times n$ biostochastic matrices, then $\mathcal{E}(B)$ is the set of permutation matrices.*

Definition 4.2.4. δ is the dirac probability measure supported at $x_0 \in \mathbb{R}^n$ if for every measurable set $A \subset \mathbb{R}^n$, we have

$$\delta_{x_0}(A) = \begin{cases} 1, x_0 \in A \\ 0, x_o \notin A \end{cases} \tag{4.2.3}$$

Theorem 4.2.5. *If* $\mu = \frac{1}{n}\sum_{i=1}^{n} \delta_{x_i}$ *and* $\nu = \frac{1}{n}\sum_{j=1}^{n} \delta_{y_j}$, *then there exists a solution to Monge's OTP.*

4.2.1 Proof of the Existence of the Discrete Monge Problem

The following proof comes from [13, §2.2].

Proof of theorem 4.2.5. Assume conditions of 4.2.5. Using our cost function, we can build an $n \times n$ matrix, where entry $(i,j) = c(x_i, y_j)$. We call this our cost matrix, and represent it by $\mathcal{C}$.

All $n \times n$ transport plans meet Kantorovich constraints. Once these plans are rescaled, we attain the set of bistochastic matrices, which we'll denote as B. We wish to minimize

$$\frac{1}{n}\Sigma_{ij}\mathcal{C}_{ij}\pi_{ij}$$

for $\pi \in B$. By compactness, we attain a minimum, which we'll call M. Pick some $\epsilon \geq 0$ and some π_ϵ such that $M \geq \Sigma_{ij}\mathcal{C}_{ij}\pi_{\epsilon,ij} - \epsilon$. Let there be affine f such that $f(\pi) = \Sigma_{ij}\mathcal{C}_{ij}\pi_{ij}$. By Minkowski-Caratherodory, we have some η supported on $\mathcal{E}$(B) such that $f(\pi_\epsilon) = \int f(\pi)\, d\eta(\pi)$. This means

$$M \geq \int \Sigma_{ij}\mathcal{C}_{ij}\pi_{ij} d\eta(\pi) - \epsilon \geq \inf_{\pi \in \mathcal{E}(B)} \Sigma_{ij}\mathcal{C}_{ij}\pi_{ij} - \epsilon \geq \inf_{\pi \in B} \Sigma_{ij}\mathcal{C}_{ij}\pi_{ij} - \epsilon = M - \epsilon$$

Since our choice of ϵ was arbitrary, we have $\inf_{\pi \in \mathcal{E}(B)} f(\pi) = M$. This means we can find the Kantorovich minimum with extreme points. $\mathcal{E}$(B) is closed since B is closed, which means this minimum is attained at an extreme point. By Birkoff's theorem, extreme points of bistochastic matrices are permutation matrices. Let us define transport plan $T^t(x_t) = y_{\sigma^t(i)}$. This means we have a way to turn all permutation matrices into transport plans.

For any transport map T, we can define matrix π by $\pi_{ij} = \delta_{y_i = T(x_i)}$. For example, this means row i has a 1 in the column that T sends x_i to. Since T is a transport map, we know $\pi \in B$. Note that π^t is the Kantorovich minimum, and it corresponds to T^t. Then we see

$$\Sigma_{i=1}^n c(x_i, T(x_i)) = \Sigma_{ij} c_{ij} \pi_{ij} \geq \Sigma_{ij} c_{ij} \pi_{ij}^t = \Sigma_{i=1}^n c(x_i, T^t(x_i))$$

This means that T^t is the optimal Monge solution. □

4.2.2 Discrete Example

Suppose we instead have a discrete transport problem. That is, our input and target ranges, X & Y respectively, each contain a finite number of elements. Specifically, let us say that both contain 4 elements. Suppose $X = \{1, 2, 3, 4\}$ and $Y = \{10, 20, 30, 40\}$, and x_i and y_i refer to the ith element of each set. Further, let us define μ and ν respectively as $\mu = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ and $\nu = \frac{1}{n} \sum_{i=1}^n \delta_{y_i}$. These are identical uniform distributions.

Note that $T(x_i) = y_{\sigma_i}$ is the trivial transport map. In fact, any bijective function from $X \to Y$ will be, by definition, a transport map. Note that every $y_i \in Y$ must receive exactly $\frac{1}{4}$ of mass from a single $x_i \in X$. This means that all transport maps in this setting must be bijections. Since both distributions are identical, we can say that bijection between them is a permutation. This means all permutations of X are transport maps to Y. With this knowledge, we can use trial and error to find a permutation of X that gives us the Monge solution for any cost function. Suppose our cost function is $c(x, y) = |x - y|$. Then we find the permutation $\sigma = (1, 2, 3, 4)$, where $x_i \to y_i$ produces the optimal transport map, with a total cost of 90.

Suppose we also wish to find the Kantorovich solution. For every $\pi \in \Pi(\mu, \nu)$, note that $\sum_{j=1}^{4} \pi_{ij} = \frac{1}{4}$ for every i, and that $\sum_{i=1}^{4} \pi_{ij} = \frac{1}{4}$ for every j. For every $\pi \in \Pi(\mu, \nu)$, define a new matrix, $\tilde{\pi}$, by $\tilde{\pi}_{ij} = 4\pi_{ij}$ for every i and j. Now note that $\sum_{j=1}^{4} \tilde{\pi}_{ij} = 1$ for every i, and that $\sum_{i=1}^{4} \tilde{\pi}_{ij} = 1$ for every j. Since every $\pi_{ij} \geq 0$, we have $\tilde{\pi}_{ij} \geq 0$. This is exactly the set of 4×4 biostochastic matrices,

$$\mathcal{B} = \{\tilde{\pi} \in \mathbb{R}^{4\times4} : \tilde{\pi}_{ij} \geq 0, \sum_j \tilde{\pi}_{ij} = 1, \sum_i \tilde{\pi}_{ij} = 1\}$$

This means all possible transport plans can be found with the set $\mathcal{B}$. By Birkoff's theorem, the extreme points of $\mathcal{B}$, $\mathcal{E}(\mathcal{B})$, are exactly the permutation matrices of $\mathcal{B}$.The minimum will be the corresponding π to one of the $\tilde{\pi} \in \mathcal{E}(\mathcal{B})$. Like with the Monge solution, we then just need to check all permutations to see which produces the lowest total cost. This also ends up being the trivial transport plan, with cost 90, as expected.

Chapter 5

Duality

Our attempts to solve the optimal transport problem so far have involved finding a method of transportation that produces the minimal total cost. This is one approach. Duality is another.

Consider the pile of sand that we wish to move to a hole in the ground again. You charge customers $c(x, y)$ to move sand from x to y. The cost to you for this is found by two separate fees: the cost of taking sand from x, $\varphi(x)$, and the cost of moving sand to y, $\psi(y)$. Clearly, you would not want the cost, $\varphi(x) + \psi(y)$, to exceed what you charge, $c(x, y)$. Otherwise, you would be losing money. Similarly, the customer does not want to pay more than the transportation costs, as they would also be losing money by not doing it themselves.

In other words, the optimal pricing is exactly the optimal transport cost. Instead of trying to find a way to move mass that minimizes cost, we are trying to find ways to move mass that do not create a deficit for either party.

5.1 The Dual

The following theorems and definitions come from [13, §3.1].

Definition 5.1.1. Let $\mu \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Y)$ where X,Y are Polish spaces. Let $c : X \times Y \to [0, +\infty]$ be a lower semi-continuous cost function. Then we have the set of **potential functions** defined as

$$\Phi_c = \{(\varphi, \psi) \in L^1(\mu) \times L^1(\nu) : \varphi(x) + \psi(y) \leq c(x, y)\} \tag{5.1.1}$$

where the inequality holds for almost every $x \in X$ and $y \in Y$.

Definition 5.1.2. Both OTP formulations already presented (Monge and Kantorovich) are both called the **primal problems**. We seek to minimize a primal problem. Let all variables be as defined in 5.1.1. $\mathbb{J}$ is the **dual problem**, where

$$\max \mathbb{J} : L^1(\mu) \times L^1(\nu) \to \mathbb{R} \quad \mathbb{J}(\varphi, \psi) = \int_X \varphi d\mu + \int_Y \psi d\nu \tag{5.1.2}$$

Both the primal and dual problems are subject to constraints.

Theorem 5.1.3 (Kantorovich Duality)**.** *Let $\mathbb{K}$ be the primal, and $\mathbb{J}$ be the dual as defined in 5.1.2. We have*

$$\min_{\pi \in \Pi} \mathbb{K}(\pi) = \max_{(\varphi, \psi) \in \Phi_c} \mathbb{J}(\varphi, \psi) \tag{5.1.3}$$

The following definition and theorem come from [13, §3.2].

Definition 5.1.4. Let E be a normed vector space, and $\Theta : E \to \mathbb{R} \cup \{+\infty\}$ be a convex function. The **Legendre-Fenchel transform** is defined as follows

$$\Theta^* : E^* \to \mathbb{R} \cup \{+\infty\}, \quad \Theta^*(z^*) = \max_{z \in E}(\langle z^*, z \rangle - \Theta(z)) \tag{5.1.4}$$

Theorem 5.1.5 (Fenchel-Rockafellar)**.** *Let E be a normed vector space, and $\Theta, \gamma : E \to \mathbb{R} \cup \{+\infty\}$ be convex functions. Let there be $z_o \in E$ such that $\Theta(z_0), \gamma(z_o) < \infty$ and Θ is continuous at z_0. Then*

$$\inf_E(\Phi + \gamma) = \max_{z \in E}(-\Theta^*(-z^*) - \gamma^*(z^*)) \tag{5.1.5}$$

where the suprememum of the right-hand side is attained

Theorem 5.1.6. *Let $\mu \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Y)$ where X, Y are Polish spaces. Let $c : X \times Y \to [0, +\infty]$ be a lower semi-continuous cost function. Assume there exists $c_X \in L^1(\mu), c_Y \in L^1(\nu)$ such that $c(x, y) \leq c_X(x) + c_Y(y)$ almost everywhere. Also assume that*

$$M = \int_X c_X(x) d\mu(x) + \int_Y c_Y(y) d\nu(y) < \infty \tag{5.1.6}$$

Then there exists $(\varphi, \psi) \in \Phi_c$ such that

$$\max_{\Phi_c} \mathbb{J} = \mathbb{J}(\varphi, \psi) \tag{5.1.7}$$

Remark 5.1.7. Note that 5.1.6 showing the existence of a maximum to the dual problem is the dual equivalent of 3.3.3, which states that a minimum to the primal problem (Kantorovich formulation) exists.

5.2 Proof of the Duality Theorem

The following proof comes from [13, §3.3]

Proof of theorem 5.1.3. First, let us prove that

$$\sup_{\varphi, \psi \in \Phi_c} \mathbb{J} \leq \inf_{\pi \in \Pi} \mathbb{K}$$

Pick any $\varphi, \psi \in \Phi_c, \pi \in \Pi$. Note that $\varphi(x) + \psi(y) \leq c(x, y)$ for almost every x, y. Because this inequality holds for μ-almost everywhere and ν-almost everywhere, we have full measures sets A, B where $\mu(A) = 1, \nu(B) = 1$ and

$$\varphi(x) + \psi(y) \leq c(x, y) \forall (x, y) \in \mu(A) \times \nu(B)$$

Note that π has marginals μ, ν. We then have,

$$\pi(A^c \times B^c) \leq \pi(A^c \times Y) + \pi(X \times B^c) = A^c + B^c = 0$$

Now integrate this against π. We get

$$\int_X \varphi(x) d\mu + \int_Y \psi(y) d\nu = \int_{X \times Y} (\varphi(x) + \psi(y)) d\pi(x, y) \leq \int_{X \times Y} c(x, y) d\pi(x, y)$$

Since this is for every admissible dual pair in Φ_c and every transport plan in Π, we can optimize both sides of this inequality to get our intended result, i.e. that $\sup \mathbb{J}_{\varphi, \psi \in \Phi_c} \leq \inf_{\pi \in \Pi} \mathbb{K}$.

Now we wish to prove

$$\sup \mathbb{J} \geq \inf \mathbb{K}$$

Assume X, Y are compact, and c is continuous. Let $E = C_b^0(X \times Y)$ be the space of continuous bounded functions on $X \times Y$. Define the following two convex functionals on E

$$\Theta(u) = \begin{cases} 0, & u(x, y) \geq -c(x, y) \qquad \forall x, y \\ +\infty, & otherwise \end{cases}$$

$$\gamma(u) = \begin{cases} \int_X \phi d\mu + \int_Y \psi d\nu, & u(x, y) = \phi(x) + \psi(y) \\ +\infty, & otherwise \end{cases}$$

We can invoke 5.1.5 if we can identify both sides. This equation can be called the Fenchel duality. For $\Theta + \gamma$ to be definite, we need both conditions to hold. The constraint is then

$$\phi(x) + \psi(y) \geq -c(x, y),$$

and we consider the minimization problem

$$\inf_{\phi,\psi} \left\{ \int_X \phi \, d\mu + \int_Y \psi \, d\nu : \ \phi(x) + \psi(y) \geq -c(x, y) \right\}.$$

Perform a change of variables where $\tilde{\phi} = -\phi$ and $\tilde{\psi} = -\psi$. This means we can rewrite the objective function to

$$-\max \int_X \tilde{\phi} d\mu + \int_Y \tilde{\psi} d\nu$$

Note that the equation we take the supremum of is exactly $\mathbb{J}$. The constraint applied to it is exactly the constraint used to define Φ_c. That means that this term is equal to $\max_{\Phi_c} \mathbb{J}$. So the left side of the Fenchel duality equation is the negative of the OT dual problem.

Let Θ^* be the Legendre-Fenchel transform of Θ. Note that $\Theta^*(u) = 0$ when $u \geq -c$. Let λ be any $\lambda \in E^*$, where $E^* = \{f : E \to \mathbb{R}$ such that f is a bounded and linear functional$\}$. By definition, this means

$$\Theta^*(\lambda) = \max_{u \in E}(\langle \lambda, u \rangle - \Theta(u)) = \max_{u \in E}(\langle \lambda, u \rangle)$$

Applied to $-\pi$, we have $\Theta^*(-\pi) = \max_{u \geq -c}(-\int u d\pi)$. If π is a positive measure, then we achieve maximum at $u = -c$. For anywhere π is not positive, the supremum is when $u = +\infty$. Finiteness forces $\pi \geq 0$. So

$$\Theta^*(-\pi) = \begin{cases} \int c d\pi, & x \in M_+(X \times Y) \\ +\infty, & otherwise \end{cases}$$

$\gamma(\pi)$ is defined only when $u = \phi(x) + \psi(y)$, so

$$\gamma^*(\pi) = \max_{\phi,\psi}(\int_{X\times Y}(\phi(x) + \psi(y))d\pi - \int_X \phi d\mu - \int_Y \psi d\nu)$$

Recall the marginal definitions, that is $\int_{X\times Y} \phi(x)d\mu = \int_X \phi d(P_{X\#}\pi)$ and $\int_{X\times Y} \psi(y)d\nu = \int_Y \psi d(P_{Y\#}\pi)$. So we can rewrite γ^* to

$$\gamma^*(\pi) = \max_{\phi,\psi}(\int_X \phi d(P_{X\#}\pi - \mu) + (\int_Y \psi d(P_{Y\#}\pi - \nu)$$

If $P_{X\#}\pi = \mu$ and $P_{Y\#}\pi = \nu$, then the supremum is 0. Otherwise, its $+\infty$, as if the marginals are wrong, then ϕ, ψ could be scaled to blow up the value. Therefore,

$$\gamma^*(\pi) = \begin{cases} 0, & \text{if } \pi \in \Pi + \\ +\infty, & otherwise \end{cases}$$

With this, we find that the right side of the Fenchel duality is $-\inf_{\Pi} \mathbb{K}$. That means

$$-\inf_{\varphi,\psi\in\Phi} \mathbb{J} = -\sup_{\pi\in\Pi} \mathbb{K}$$

The supremum (sup) of a set is the least upper bound of a set, while the infimum (inf) is the greatest lower bound. The supremum is equal to the maximum of a set if a maximum is attained by that set. The same is true for the infimum and the minimum. By 5.1.7 we recall that both are attained. Therefore, we have proved that $\max \mathbb{J} = \min \mathbb{K}$. $\square$

Remark 5.2.1. This proof holds when X, Y are not compact, and c is only lower semi-continuous. However, the additional work to prove these claims exceeds what is expected for an undergraduate thesis.

5.3 Duality Example

Suppose we have two supply locations and two demand locations represented by

Location	Mass
x_1	0.5
x_2	0.5

Table 5.1: Demand Table

Location	Mass
y_1	0.5
y_2	0.5

Table 5.2: Supply Table

and cost matrix

$$\begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}$$

Trivially, the optimal transport plan is

$$\begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix}$$

which produces total cost of 1. In other words, $\min \mathbb{K} = 1$. This is the primal solution. Now let us solve it as a dual problem.

The dual variables ϕ, ψ must satisfy $\phi(x) + \psi(y) \leq c(x, y)$ for every x, y pair. The dual objective has the form

$$\mathbb{J}(\phi, \psi) = \sum_x \phi(x)\mu(x) + \sum_y \psi(y)\nu(y) = 0.5(\phi_1 + \psi_1 + \phi_2 + \psi_2)$$

We wish to maximize this objective. Let us note our constraints.

$$\begin{aligned} \phi_1 + \psi_1 \leq 1, \quad \phi_1 + \psi_2 \leq 3 \\ \phi_2 + \psi_1 \leq 2, \quad \phi_2 + \psi_2 \leq 1 \end{aligned}$$

Note the strictest constraints mean that $\phi_1 + \psi_1 + \phi_2 + \psi_2 \leq 2$. We can get many feasible solutions from this. Let us suppose that $\phi_1 = 1, \psi_1 = 0, \phi_2 = 1,$ and $\psi_2 = 0$.

There are four possible ways to move mass during transport, each of which is represented by an entry in the cost matrix. Since we are looking for our optimum, and it occurs when $\phi_i + \psi_j = c_{ij}$ for all i, j, then we have to compare dual pairs to the corresponding cost to check if that path is optimal. We see that only routes $(1,1)$ and $(2,2)$ meet the equality. Therefore, we know the optimal transport plan will have the form

$$\begin{pmatrix} ?_1 & 0 \\ 0 & ?_2 \end{pmatrix}$$

By checking intended row and column sums, we see that $?_1 = ?_2 = 0.5$, which gives us the transport plan from before.

Chapter 6

General Solutions

We have covered several ways to solve some OTPs. However, these problems were limited in size and complexity. Most OTPs involve multi-dimensional probability distributions, non-trivial cost functions, and data sizes beyond basic human intuition. More applicable methods are needed. Two of these methods are outlined in this chapter.

6.1 Linear Programming

Definition 6.1.1. [8, §3.1]. **Linear programming** is an optimization method for linear problems. That means the function we'd like to optimize the output of, the objective function, is linear, as well as all constraints must be linear equalities or inequalities.

Both the primal and the dual OTPs fit this description exactly. They can be turned into linear programs (LPs), and then solved.

The following remarks come from [8, §2.3]

Remark 6.1.2. If either the source or target distribution is continuous, then we would be attempting to solve an infinite-dimensional linear program (LP). Discretizing continuous distributions gives an approximate finite-dimensional problem that is easier to solve. Therefore, we proceed by discretizing all continuous distributions.

Remark 6.1.3. The set of feasible solutions to Monge problems are nonconvex, which makes solutions hard to find. Therefore, we focus only on Kantorovich problems. This also guarantees solutions, which Monge problems often do not have.

The following results come from [8, §2.3]

The primal OTP from $\mu \in \mathbb{R}^n$ to $\nu \in \mathbb{R}^m$ can be expressed as

$$\mathbb{K}(\pi) = \sum_{i}^{n} \sum_{j}^{m} c_{ij}\pi_{ij} \tag{6.1.1}$$

where we want to find the $\pi \in \Pi(\mu, \nu)$ that minimizes $\mathbb{K}(\pi)$. This solution will be denoted as π^*. Let us set this into typical LP format.

Let $\otimes$ be the **Kronecker product**, where each entry of the first matrix multiplies the entire second matrix to create a larger matrix.

Example 6.1.4.

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 5 \\ 6 & 7 \end{pmatrix}$$

$$A \otimes B = \begin{pmatrix} 1 \cdot B & 2 \cdot B \\ 3 \cdot B & 4 \cdot B \end{pmatrix}$$

$$A \otimes B = \begin{pmatrix} 0 & 5 & 0 & 10 \\ 6 & 7 & 12 & 14 \\ 0 & 15 & 0 & 20 \\ 18 & 21 & 24 & 28 \end{pmatrix}$$

Let $\mathbb{I}_n$ be the **identity matrix** of size $n \times n$, and 1_n a size n matrix of with **entries all 1s**. Now we define the following matrix

$$A = \begin{pmatrix} 1_n^T \otimes \mathbb{I}_m \\ \mathbb{I}_n \otimes 1_m^T \end{pmatrix} \tag{6.1.2}$$

where $A \in \mathbb{R}^{(n+m)\times nm}$. Now define **vector** $p \in \mathbb{R}^{nm}$ by by flattening the transport plan π with the rule that $p_{i+n(j-1)} = \pi_{ij}$. Further, let $\mathcal{C}$ be the **cost matrix** where $\mathcal{C}_{ij} = c(x_i, y_j)$. We wish to find

$$\min \mathbb{K}(p) = \mathcal{C}^T p : Ap = \begin{pmatrix} \mu \\ \nu \end{pmatrix} = b \in \mathbb{R}^{n+m} \tag{6.1.3}$$

The dual is subsequently found as

$$\max \mathbb{J} = b^T h \text{ such that } A^T h \leq \mathcal{C} \text{ and } h = \begin{pmatrix} f \\ g \end{pmatrix} \tag{6.1.4}$$

where $f \in \mathbb{R}^n, g \in \mathbb{R}^m$. Here, f and g are the **vectorized dual potentials**.

6.1.1 The Simplex Algorithm

The following description of the simplex algorithm is adapted from [14, §2].

Definition 6.1.5. The **simplex algorithm** is a method for solving linear optimization problems. It proceeds by moving between **basic feasible solutions** (BFS), improving the objective value at each step until an optimal solution is reached.

In the context of optimal transport problems (OTPs), the simplex algorithm operates on transport plans and exploits the structure of the constraint set. While the method can be applied to both the primal and dual problems, we describe it here for the primal formulation.

We now outline the steps of the simplex algorithm for the primal optimal transport problem.

1. **Construct an initial basic feasible solution (BFS).**

   We seek a feasible transport plan π satisfying the marginal constraints. A standard way to construct such a plan is the **northwest corner rule**:

   - Start at the entry $(1, 1)$ and assign

   $$\pi_{11} = \min\{\text{row sum}_1, \text{column sum}_1\}.$$

   - If the first row is not yet satisfied, move to $(1, 2)$; if the first column is not yet satisfied, move to $(2, 1)$.
   - Continue this process, moving right or down depending on which marginal constraint is not yet met.
   - Repeat until all row and column sums are satisfied.

   This produces a feasible transport plan π with at most $n + m - 1$ nonzero entries, which forms a BFS.

2. **Determine the basis and compute simplex multipliers.**

The basic variables correspond to the nonzero entries of π, chosen so that they form a spanning tree structure (i.e., no cycles and exactly $n + m - 1$ basic variables).

We introduce **simplex multipliers** (dual variables)

$$\alpha_i \quad (i = 1, \ldots, n), \qquad \beta_j \quad (j = 1, \ldots, m),$$

which satisfy

$$\alpha_i + \beta_j = c_{ij} \quad \text{for all basic cells } (i, j).$$

To determine these values:

- Fix one multiplier arbitrarily (e.g., set $\alpha_1 = 0$),
- Solve the resulting system of equations over all basic cells.

3. **Compute reduced costs and test for optimality.**

For each non-basic cell (i, j), compute the **reduced cost**

$$\bar{c}_{ij} = c_{ij} - (\alpha_i + \beta_j).$$

- If all reduced costs satisfy $\bar{c}_{ij} \geq 0$, then the current transport plan π is optimal.
- Otherwise, select a non-basic cell with $\bar{c}_{ij} < 0$ to enter the basis.

4. **Pivot to improve the solution.**

Adding the chosen entering cell to the basis creates a unique cycle in the grid formed by the basic variables.

- Trace this cycle by alternating horizontal and vertical moves through basic cells.

- Assign alternating $+$ and $-$ signs along the cycle, starting with $+$ at the entering cell.

- Let
$$\theta = \min\{\pi_{ij} : (i,j) \text{ is marked with } -\}.$$

- Update the transport plan:
$$\pi_{ij} \leftarrow \begin{cases} \pi_{ij} + \theta & \text{if } (i,j) \text{ has a } +, \\ \pi_{ij} - \theta & \text{if } (i,j) \text{ has a } -. \end{cases}$$

One basic variable becomes zero and leaves the basis, preserving the number of basic variables.

5. **Repeat.**

Return to Step 2 and repeat the process until all reduced costs are nonnegative. The resulting transport plan π^* is optimal.

6.1.2 Linear Programming Example

Let μ be the uniform distribution on the triangle with vertices $(0,0)$, $(1,0)$, and $(0,1)$, and let ν be the uniform distribution on the line segment with endpoints $(1,0)$ and $(1,1)$. We want to transport μ to ν. This optimal transport problem can be approximated by a finite-dimensional linear program and solved with the simplex algorithm.

Step 1: Discretize the measures.

We divide each distribution into three pieces of equal mass and represent each piece by its centroid or midpoint. Each piece is assigned mass $\frac{1}{3}$. For μ, we use the points $x_1 = (0.25, 0.25), x_2 = (0.6, 0.2), x_3 = (0.2, 0.6)$ and for ν, we use $y_1 = \left(1, \frac{1}{6}\right), y_2 =$

$\left(1, \frac{1}{2}\right), y_3 = \left(1, \frac{5}{6}\right)$. Thus, a transport plan is a 3×3 matrix

$$\pi = \begin{pmatrix} \pi_{11} & \pi_{12} & \pi_{13} \\ \pi_{21} & \pi_{22} & \pi_{23} \\ \pi_{31} & \pi_{32} & \pi_{33} \end{pmatrix},$$

where each row sum and each column sum must equal $\frac{1}{3}$. We take the cost function to be the Euclidean distance

$$c(x, y) = \sqrt{(x^{(1)} - y^{(1)})^2 + (x^{(2)} - y^{(2)})^2}.$$

This gives the cost matrix

$$\mathcal{C} = \begin{pmatrix} 0.585 & 0.625 & 0.751 \\ 0.203 & 0.290 & 0.471 \\ 0.938 & 0.650 & 0.583 \end{pmatrix}.$$

If we flatten π into a vector $p = (\pi_{11}, \pi_{12}, \pi_{13}, \pi_{21}, \pi_{22}, \pi_{23}, \pi_{31}, \pi_{32}, \pi_{33})^T$ then the linear program is $\min \mathbb{K}(p) = \mathcal{C}^T p$ subject to $Ap = b$ and $p \geq 0$ where A encodes the row and column sum constraints and $b = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)^T$.

Step 2: Find an initial basic feasible solution.

We construct an initial feasible transport plan using the northwest corner rule. Start at $(1, 1)$. Since both the first row and first column need total mass $\frac{1}{3}$, we assign $\pi_{11} = \frac{1}{3}$. This completely satisfies row 1 and column 1, so all other entries in that row and column must be zero. At this stage,

$$\pi^{(0)} = \begin{pmatrix} \frac{1}{3} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Next move diagonally to $(2, 2)$, and assign $\pi_{22} = \frac{1}{3}$. Again, this completely satisfies row 2 and column 2, so the remaining entries in that row and column must be zero. Now

$$\pi^{(1)} = \begin{pmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Finally, move to $(3, 3)$, and assign $\pi_{33} = \frac{1}{3}$. This gives

$$\pi^{(2)} = \begin{pmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{3} \end{pmatrix}.$$

This plan satisfies all row and column constraints, so it is a basic feasible solution.

Step 3: Choose a basis and compute the simplex multipliers.

A basis for a 3×3 transport problem must contain $n + m - 1 = 5$ variables. We take the three positive entries of $\pi^{(2)}$, together with two adjacent zero entries, namely

$$(1, 1),\ (1, 2),\ (2, 2),\ (2, 3),\ (3, 3).$$

These basis cells form a connected path with no cycle. Now introduce simplex multipliers $\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, \beta_3$ which satisfy $\alpha_i + \beta_j = c_{ij}$ for each basic cell (i, j).

So we have

$$\alpha_1 + \beta_1 = 0.585, \qquad \alpha_1 + \beta_2 = 0.625, \qquad \alpha_2 + \beta_2 = 0.290,$$

$$\alpha_2 + \beta_3 = 0.471, \qquad \alpha_3 + \beta_3 = 0.583.$$

Set $\alpha_1 = 0$. Solving gives

$$\alpha = (0, -0.335, 0.112), \qquad \beta = (0.585, 0.625, 0.806).$$

Step 4: Compute reduced costs.

For every non-basic cell, compute the reduced cost $\bar{c}_{ij} = c_{ij} - (\alpha_i + \beta_j)$. The non-basic cells are $(1,3)$, $(2,1)$, $(3,1)$, and $(3,2)$. Their reduced costs are

$$\bar{c}_{13} = 0.751 - (0 + 0.806) = -0.055,$$

$$\bar{c}_{21} = 0.203 - (-0.335 + 0.585) = -0.047,$$

$$\bar{c}_{31} = 0.938 - (0.112 + 0.585) = 0.241,$$

$$\bar{c}_{32} = 0.650 - (0.112 + 0.625) = -0.087.$$

Since some reduced costs are negative, the current plan is not optimal. The most negative reduced cost is at $(3,2)$, so this cell enters the basis.

Step 5: Pivot and update π.

Adding $(3,2)$ to the basis creates the cycle

$$(3,2)^+ \to (2,2)^- \to (2,3)^+ \to (3,3)^- \to (3,2).$$

However, because $(2,3)$ is currently a basic zero cell, this pivot does not yet improve the solution in a useful way. So we instead use the negative reduced cost at $(2,1)$, which leads to a nondegenerate pivot. Adding $(2,1)$ creates the cycle

$$(2,1)^+ \to (1,1)^- \to (1,2)^+ \to (2,2)^- \to (2,1).$$

We now alternate $+$ and $-$ signs around the cycle. The cells marked with $-$ are $(1,1)$ and $(2,2)$, both of which currently contain mass $\frac{1}{3}$. Hence $\theta = \min\{\pi_{11}, \pi_{22}\} = \frac{1}{3}$. Now update the entries along the cycle:

$$\pi_{21} \leftarrow \pi_{21} + \theta = 0 + \frac{1}{3} = \frac{1}{3},$$

$$\begin{aligned}
\pi_{11} &\leftarrow \pi_{11} - \theta = \frac{1}{3} - \frac{1}{3} = 0, \\
\pi_{12} &\leftarrow \pi_{12} + \theta = 0 + \frac{1}{3} = \frac{1}{3}, \\
\pi_{22} &\leftarrow \pi_{22} - \theta = \frac{1}{3} - \frac{1}{3} = 0.
\end{aligned}$$

So the transport plan becomes

$$\pi^{(3)} = \begin{pmatrix} 0 & \frac{1}{3} & 0 \\ \frac{1}{3} & 0 & 0 \\ 0 & 0 & \frac{1}{3} \end{pmatrix}.$$

Step 6: Check optimality.

Using the new basis, we recompute the simplex multipliers and then the reduced costs. We find that all non-basic reduced costs are now nonnegative. Therefore, $\pi^{(3)}$

is optimal. Hence $\pi^* = \pi^{(3)}$ and its total transport cost is

$$\mathbb{K}(\pi^*) = \frac{1}{3}(0.625) + \frac{1}{3}(0.203) + \frac{1}{3}(0.583) \approx 0.470.$$

6.2 Regularized Optimal Transport

Linear programming can be used to find the exact solution of any solvable OTP. Despite this fact, it is often not used in practice. Its worst-case running time is $O(2^n)$, which is not scalable. Many find it preferable to trade a slow but exact solution for a quick approximate one. This is where regularization comes into play.

The following definitions come from [8, §4.1].

Definition 6.2.1. The **Entropy** of matrix π is defined as

$$H(\pi) = -\Sigma_{ij}\pi_{ij}(log(\pi_{ij}) - 1) \tag{6.2.1}$$

Recall our discrete Kantorovich objective:

$$\min_{\pi \in \Pi} \mathbb{K}(\pi) = \Sigma_{ij}\mathcal{C}_{ij}\pi_{ij} = \langle \mathcal{C}, \pi \rangle$$

Definition 6.2.2. With an entropy term, we can make the Kantorovich problem into a **regularized primal problem**. For some $\epsilon > 0$ and a discrete Kantorovich problem,

$$\min_{\pi \in \Pi} \mathbb{K}_\epsilon(\pi) = \Sigma_{ij}\mathcal{C}_{ij}\pi_{ij} - \epsilon H(\pi) = \langle \mathcal{C}, \pi \rangle - \epsilon H(\pi) \tag{6.2.2}$$

Denote the solution to this problem as π_ϵ^*. Intuitively, as $\epsilon \to 0$, $\pi_\epsilon^* \to \pi^*$. This means any solution to $\mathbb{K}_\epsilon$ will be an approximate solution of $\mathbb{K}$.

Remark 6.2.3. The objective function in 6.2.2 is given by $\langle \mathcal{C}, \pi \rangle - \epsilon H(\pi)$. The term $\langle \mathcal{C}, \pi \rangle$ is linear in π, and therefore convex. The entropy term $H(\pi)$ is strictly concave, since its Hessian is given by $\nabla^2 H(\pi) = -\operatorname{diag}\left(\frac{1}{\pi_{ij}}\right)$ which is negative definite on the interior of the domain. The Hessian is a square matrix of second-order partial derivatives of $\langle \mathcal{C}, \pi \rangle$ with respect to π.It follows that $-\epsilon H(\pi)$ is strictly convex.

Moreover, if $\pi_{ij} \geq \delta > 0$ for all i, j, then the Hessian is bounded below, and $-\epsilon H(\pi)$ is $\frac{\epsilon}{\delta}$-strongly convex. Since the sum of a convex function and a (strictly or strongly) convex function is again (strictly or strongly) convex, it follows that the objective in 6.2.2 is strictly convex, and in fact strongly convex on the interior of its domain.

Note that $\nabla^2 H(\pi)$ is infinitely differentiable, making 6.2.2 by theorem 2.2.4 smooth.

Remark 6.2.4. The feasible set $\Pi(\mu, \nu)$ is convex. Indeed, if $\pi_1, \pi_2 \in \Pi(\mu, \nu)$ and $\lambda \in [0, 1]$, then $\lambda\pi_1 + (1 - \lambda)\pi_2$ satisfies the same marginal constraints and remains non-negative, so it also belongs to $\Pi(\mu, \nu)$.

Furthermore, every entry of $\pi \in \Pi(\mu, \nu)$ satisfies $0 \leq \pi_{ij} \leq \min(\mu_i, \nu_j)$, so $\Pi(\mu, \nu)$ is bounded. The marginal and non-negativity constraints are closed, hence $\Pi(\mu, \nu)$ is closed. Therefore, $\Pi(\mu, \nu)$ is compact.

Since the objective function in 6.2.2 is continuous, it follows from the extreme value theorem that a minimizer exists on $\Pi(\mu, \nu)$.

Remark 6.2.5. We now show that the minimizer is unique and is characterized by the Lagrange optimality conditions.

Since the objective function in 6.2.2 is convex and the feasible set $\Pi(\mu, \nu)$ is convex, this defines a convex optimization problem. Let $\pi^* \in \Pi(\mu, \nu)$ satisfy the Lagrange equations.

Take any $\pi \in \Pi(\mu, \nu)$ and consider the line $\gamma(t) = \pi^* + t(\pi - \pi^*)$ which lies entirely in $\Pi(\mu, \nu)$ by convexity. Define $g(t) = K(\gamma(t))$ where K is the objective function. Since K is convex, g is a convex function of t. The Lagrange conditions imply that $g'(0) = 0$, and by strict convexity of g, it follows that $g(0) < g(t)$ for all $t \neq 0$. In particular, taking $t = 1$, we obtain $K(\pi^*) = g(0) < g(1) = K(\pi)$.

Since $\pi \in \Pi(\mu, \nu)$ was arbitrary, this shows that π^* is the unique global minimizer. Therefore, 6.2.2 admits a unique solution on $\Pi(\mu, \nu)$.

[8, §4.1] points out that the unique minimum means that convergence algorithms will be more stable and provide more diffused results. For this, and more reasons to be given, regularized OT is heavily favored to exact methods, like linear programming.

The dual is given in [8, §4.4]

Definition 6.2.6. The **regularized dual** is given by

$$\max_{f,g\in\Phi} \mathbb{J}(f,g) = \langle f,\mu\rangle + \langle g,\nu\rangle - \epsilon\langle e^{f/\epsilon}, \mathcal{K}e^{g/\epsilon}\rangle \tag{6.2.3}$$

where $(u,v) = (e^{f/\epsilon}, e^{g/\epsilon})$, $f_i = \mu_i$, and $g_j = \nu_j$ and all other variables are as given before.

Definition 6.2.7. The **regularized dual problem** is given by

$$\max_{f\in\mathbb{R}^n,\, g\in\mathbb{R}^m} \mathbb{J}(f,g) = \langle f,\mu\rangle + \langle g,\nu\rangle - \epsilon\left\langle e^{f/\epsilon}, \mathcal{K}e^{g/\epsilon}\right\rangle, \tag{6.2.4}$$

where $\mathcal{K}_{ij} = e^{-C_{ij}/\epsilon}$, the exponential is taken componentwise, and f,g are our dual potentials,

Remark 6.2.8. $\mathbb{J}^{'}(f,g) \approx e^{f,g/\epsilon}$, so 6.2.7 is smooth, and strongly concave. This means it also has linear convergence, and if it has a local maximum, it is both unique and the global maximum.

Remark 6.2.9. Note that if $f,g\to\infty$, then $-\epsilon\langle e^{f/\epsilon}, \mathcal{K}e^{g/\epsilon}\rangle \to -\infty$ much faster than $\langle f,\mu\rangle + \langle g,\nu\rangle \to \infty$. Similarly note that if $f,g\to-\infty$, then $-\epsilon\langle e^{f/\epsilon}, \mathcal{K}e^{g/\epsilon}\rangle \to 0$ and $\langle f,\mu\rangle + \langle g,\nu\rangle \to -\infty$. This means that for any $c\in\mathbb{R}$, the closed ball

$$\{(f,g) : \mathbb{J}(f,g) \geq c\} \tag{6.2.5}$$

is bounded. Since $\mathbb{J}$ is continuous, it attains some maximum on any non-empty ball.

These remarks similarly show that the regularized dual gives fast and stable convergence.

6.2.1 The Sinkhorn Algorithm

There are multiple ways to solve a regularized OTP. The most common one is the Sinkhorn algorithm. [8, §4.2] finds its running time to be $O(n^2 log(n))$, which makes it significantly faster than linear programming. It is still more efficient than most other regularized algorithms, as each iteration of the Sinkhorn algorithm only requires two matrix-vector multiplications and optimizing over two vectors, one of size $\mathbb{R}^n$ and one of size $\mathbb{R}^m$. Other methods, like the gradient method outlined in [7, §2.1.5] involve calculating several gradients of size $\mathbb{R}^{n\times m}$, which is much more expensive.

The following results are from [8, §4.2].

Theorem 6.2.10. *The solution to 6.2.2 will have the form*

$$\pi_\epsilon^* = diag(u)\mathcal{K}diag(v) \tag{6.2.6}$$

where $\mathcal{K} = e^{-\mathcal{C}/\epsilon}$*, and* u *and* v *are two scaling vectors chosen so that marginal constraints are satisfied. This means that* $\pi_\epsilon^* \times 1_m = u \odot (\mathcal{K}v) = \mu$ *and* $(\pi_\epsilon^*)^T \times 1_n = v \odot (K^T u) = \nu$.

Remark 6.2.11. The operator $\odot$ refers to the Hadamard product, which is entrywise matrix multiplication for matrices of the same dimensions.

Theorem 6.2.12. *To find* u *and* v *as in 6.2.6, we first initialize* $u^{(0)} = 1_n$ *and* $v^{(0)} = 1_m$*. Then, until convergence, we iteratively update each variable with*

$$u^{(k+1)} = \frac{\mu}{\mathcal{K}v^{(k)}} \quad and \quad v^{(k+1)} = \frac{\nu}{\mathcal{K}^T u^{(k+1)}} \tag{6.2.7}$$

Note that μ, ν, u, v are vectors, while $\mathcal{K}$ is a matrix. These equations are done componentwise.

As $\epsilon \to 0$, we get $\pi_\epsilon^* \to \pi^*$. However, issues arise in practice often before reaching a desired error. As $\epsilon \to 0$, some values of $\mathcal{K}$ become too small to be stored in memory as positive numbers and instead are saved as 0s. This is called overflow. Eventually, it may result in a division by 0 in an iteration of 6.2.7, which will cause the algorithm to fail. We can reparameterize 6.2.6 to avoid this problem.

Theorem 6.2.13. *Let $f = \epsilon log(u)$ and $g = \epsilon log(v)$ be log domain variables. With these variables, we find that*

$$\pi_{\epsilon,ij}^* = \exp\left(\frac{f_i + g_j - \mathcal{C}_{ij}}{\epsilon}\right) \tag{6.2.8}$$

where we approximate f and g by

$$f_i = \epsilon(log\mu_i - log\Sigma_j(\frac{g_j - \mathcal{C}_{ij}}{\epsilon})) \quad and \quad g_j = \epsilon(log\nu_j - log\Sigma_i(\frac{f_i - \mathcal{C}_{ij}}{\epsilon})) \tag{6.2.9}$$

and we update until convergence. Here, convergence means that the maximum euclidean difference between row sums and μ & column sums and ν is less than our epsilon value.

Log-domain variables avoid overflow by working with the logarithms of u and v. Since logarithms convert multiplications into additions and compress the range of possible values, the algorithm updates with log-sum computations rather than direct exponentials. This keeps values within floating-point limits.

Remark 6.2.14. The Sinkhorn algorithm described above produces, but does not update or optimize over transport plans. This means that it is not a primal algorithm. Instead, we are iteratively enforcing dual optimality conditions on our potentials, which we reparameterized as scaling vectors, until they converge. This makes it more like a dual algorithm. Recall that the dual's optimal coincides with the primal's optimum. Once converged, we can use the potentials to build the optimal transport plan with these resulting values.

6.2.2 Proof of the Sinkhorn Solution

Proof. The following proof of 6.2.10 comes from [8, §4.2].

First, let us rewrite our constraints as vectors f, g where $f_i(\pi) = \sum_{j=1}^{m} \pi_{ij} = \mu_i$ and $g_j(\pi) = \sum_{i=1}^{n} \pi_{ij} = \nu_j$. Then, we introduce Lagrangian multipliers $\lambda \in \mathbb{R}^n$ and $\kappa \in \mathbb{R}^m$. With these, we can build the Lagrangian for 6.2.2

$$\mathcal{L}(\pi, \lambda, \kappa) = \mathbb{K}(\pi) - \sum_{i=1}^{n} \lambda_i (f_i(\pi) - \mu_i) - \sum_{j=1}^{m} \kappa_j (g_j(\pi) - \nu_j)$$

Then, we differentiate $\mathcal{L}$ with respect to π_{ij} for each i, j.

$$\frac{\partial}{\partial \pi_{ij}} \mathcal{L} = \mathcal{C}_{ij} + \epsilon log(\pi_{ij}) - \lambda_i - \kappa_j$$

Each of the above equations will equal 0 at optimality. When we solve these for corresponding transport plan entries, we then build the transport plan at optimality. In other words, we find π^*. This is expressed as

$$\pi^*_{\epsilon,ij} = e^{\lambda_i/\epsilon} e^{-\mathcal{C}_{ij}/\epsilon} e^{\kappa_j/\epsilon}$$

Let $u_i = e^{\lambda_i/\epsilon}, v_j = e^{\kappa_j/\epsilon}$, and $\mathcal{K}_{ij} = e^{-\mathcal{C}_{ij}/\epsilon}$. From this we get the result

$$\pi^*_\epsilon = diag(u) \mathcal{K} diag(v)$$

□

6.2.3 Proof of Sinkhorn Iterations

Proof. The following proof of 6.2.12 comes from

Let $K = e^{-\mathcal{C}/\epsilon}$, $A = \{\pi \geq 0 : \pi 1_m = \mu\}$ and $B = \{\pi \geq 0 : \pi^T 1_n = \nu\}$. Regularized OT is equivalent to

$$\pi^* = \arg \min_{\pi \in A \cap B} KL(\pi \| \mathcal{K})$$

where

$$KL(\pi \| \mathcal{K}) = \sum_{ij} \pi_{ij} log(\frac{\pi_{ij}}{\mathcal{K}_{ij}}) - \pi_{ij} + \mathcal{K}_{ij}$$

By 2.7.3, the sequence defined as

$$\pi^{(0)} = \mathcal{K}, \pi^{(1)} = \Pi_A(\pi^{(0)}), \pi^{(2)} = \Pi_B(\pi^{(1)})$$

will converge to π^*.

Let Q be the previous entry in the sequence $\pi^{(t)}$.

$$\Pi_A(Q) = \arg \min_{\pi \in A} KL(\pi \| Q)$$

separates by rows because the constraint $\pi \times 1 = \mu$ is row-wise (i.e. $\sum_j \pi_{ij} = \mu_i$). So for each row i, we need to solve

$$\min \sum_j (\pi_{ij} log(\frac{\pi_{ij}}{Q_{ij}}) - \pi_{ij})$$

We do this by taking the Lagrangian,

$$\mathcal{L}_i(\pi_i, \lambda_i) = \sum (\pi_{ij} log(\frac{\pi_{ij}}{Q_{ij}}) - \pi_{ij} + \lambda_i(\sum_j \pi_{ij} - \mu_i))$$

Taking the partial derivative with respect to π_{ij}, setting equal to 0, and then isolating for π_{ij}, we see $\pi_{ij} = Q_{ij}e^{-\lambda_i}$. Recall that $\sum_j \pi_{ij} = \mu_i$. Rearranging the previous equation with this sum, we see that $e^{-\lambda_i} = \frac{\mu_i}{Q_i}$, so

$$\Pi_A(Q) = diag\left(\frac{\mu_i}{Q_i}\right) Q$$

If $Q = diag(u)\mathcal{K}diag(v)$, then $\frac{\mu}{Q} = \frac{\mu}{u(\mathcal{K}v)}$. This can be rearranged to show $u = \frac{\mu}{\mathcal{K}v}$. A similar argument can be made to show $v = \frac{\nu}{\mathcal{K}^T u}$.

Start with $\mu^{(0)} = 1$ and $\nu^{(0)} = 1$. Then $\pi^{(0)} = \mu^{(0)}\mathcal{K}\nu^{(0)}$. Each iteration has $\pi^{(t)} = \mu^{(t)}\mathcal{K}\nu^{(t)}$. Subsequently, $u^{(t)} = \frac{\mu}{\mathcal{K}v^{(t)}}$ and $\nu^{(t)} = \frac{\nu}{\mathcal{K}^T u^{(t)}}$. By Bregman's theorem, $\pi^t \to \pi^*$, but since $\pi^{(t)} = diag(u^{(t)})\mathcal{K}diag(v^{(t)})$, we have $\mu^{(t)} \to \mu$ and $\nu^{(t)} \to \nu$.

$\square$

Remark 6.2.15. The Sinkhorn algorithm approximates u, v to meet marginality conditions. This means that π^*_ϵ, which depends on u, v, also approximately meets marginality conditions.

6.2.4 Sinkhorn Example

Let us reuse the set up from our linear programming example. Then, we initialize $f^{(0)} = (0,0,0)$ and $g^{(0)} = (0,0,0)$. Then, we use 6.2.13 to find $f^{(1)} = (0.510, 0.140, 0.516)$. We similarly find $g^{(1)} = (-0.021, 0.021, 0.010)$. With these we get

$$\pi^{(1)}_\epsilon = \begin{pmatrix} 0.148 & 0.153 & 0.010 \\ 0.186 & 0.075 & 0.001 \\ 0 & 0.105 & 0.322 \end{pmatrix}$$

Which gives column sums $(\pi^{(1)}_\epsilon)^T \times 1_3 = (0.333, 0.333, 0.333)$, and row sums $\pi^{(1)}_\epsilon \times 1_3 =$

$(0.311, 0.263, 0.427)$. Both are meant to be close to $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, which is true for our column sums, but not row sums. Our tolerance could be any $\epsilon > 0$, but we'll use $\epsilon = 0.01$.

We find that $f^{(8)} = (0.527, 0.169, 0.477)$ and $f^{(9)} = (-0.044, 0.013, 0.045)$, which gives

$$\pi^*_\epsilon = \begin{pmatrix} 0.125 & 0.177 & 0.027 \\ 0.208 & 0.115 & 0.006 \\ 0 & 0.041 & 0.300 \end{pmatrix}$$

Which gibes column sums $(\pi^{(1)}_\epsilon)^T \times 1_3 = (0.333, 0.333, 0.333)$ and row sums $\pi^{(1)}_\epsilon \times 1_3 = (0.330, 0.329, 0.341)$. The maximum Euclidean distance between either of these sums and our goals of $(1/3, 1/3, 1/3)$ is 0.007881, which is less than $\epsilon = 0.01$. Therefore, we have achieved convergence with the above matrix and cost 0.485. This cost is quite similar to that found with LP.

$\mathcal{C}, \pi^* \in \mathbb{R}^{n \times m}$

Chapter 7

Colour Transfer

With theory and methods to implement optimal transport, let us now solve a real life problem with optimal transport. Suppose you have an image which is very bland and gray. Suppose you have another image that is vibrant and colourful. You wish to make the first image appear more like the second.

The general description of colour transfer is from [1].

Definition 7.0.1. Colour grading is a technique using in photography and film to change the colour, contrast, or tone of an image to achieve a certain look. **Colour transfer** is a type of colour grading where you change the colours of one image to look like that of another image's. This can be solved as an optimal transport problem.

Every image is composed of a certain number of pixels. Each pixel has a colour, which can be represented numerically. For monochromatic images, this number falls in range $[0, 1]$, while chromatic images will need vectors to represent each colour. Typically, this is the RGB model, which measures how red, green, and blue a colour is with a vector in $\mathbb{R}^3$.

Once each pixel in an image has been assigned a colour value, you can then find what proportion of the image is made of each colour. This forms the image's colour distribution. This is a discrete distribution. To turn the source image's colour distribution into that of the target image's, we'll have to change the color of many pixels. However, we want to keep pixels as close as possible to their original colour. In other words, we still want to solve the overall cost function. This is a basic discrete Kantorovich OTP which we can solve with the reparameterized Sinkhorn algorithm.

Definition 7.0.2. [8, §4.2] The optimal transport plan produced for a pixel's colour will be a vector of colours. A colour cannot be mapped to several. **Barycentric Projection** averages this vector to produce a single colour output. After finding the optimal transport plan, this is done for each pixel. This creates a pseudo-transport map, where every pixel's colour is mapped to just one colour.

7.0.1 Implementation Structure

This colour transfer project was coded in python on a Jupyter notebook. It uses only five libraries, listed below

1. __future__ : used to add annotations to describe methods
2. numpy: used to work with arrays, a matrix-like structure
3. PIL: used to work with images
4. matplotlib: used to display images
5. scipy: used for pre-built mathematical functions

One practical issue regarding colour transfer is noise, which is highlighted in [1]. It is roughly expected that a region of similar colour in the source would be mapped into a region of similar colour. Yet, it may be more cost effective to map small parts of it into a very different colour. This creates noise in the output. Regularization, which creates much smoother results, helps combats this. An additional method is to include geographic location as a factor for each pixel. Pixels near each other in the source image will need to be mapped to pixels near each other in the target image.

Even a small and low resolution image can contain over 2×10^5 pixels. Optimal transport with dimensions of this scale would be incredibly costly. This needs to be made more efficient.

One approach is randomly sampling pixels from both the source and target images. The Sinkhorn algorithm is performed only between these samples. Then, once complete, the untransformed pixels in the source image are mapped to the nearest sampled pixel, then transformed into the same colour as the sample pixel. Intuitively, choosing larger sized sample sets results in more accurate and smooth results, but will also take exponentially more time.

Another method is to limit the number of Sinkhorn iterations done during transfer, even if columns/row have not yet converged. Without this limit, the algorithm could run for up to thousands of iterations. This method would reduce quality of results, but significantly improve speed.

7.0.2 Colour Transfer Code

Below is the entire code created for this colour transfer project. It is presented in blocks, each representing a function.

```
def colour_transfer(src_path: str, tgt_path: str, out_path: str,
                    eps: float, sinkhorn_iters: int,
                    sinkhorn_tol: float, check_every: int,
```

```
                    w_pos: float,
                    ns: int, nt: int, seed: int,
                    cost_chunk: int, nn_chunk: int) -> None:
    """
    Run global OT colour transfer and save output.
    """
    rng = np.random.default_rng(seed)

    src = load_rgb_255(src_path)
    tgt = load_rgb_255(tgt_path)

    Hs, Ws, _ = src.shape
    Ht, Wt, _ = tgt.shape
    print(f"Source: {Hs}x{Ws}, Target: {Ht}x{Wt}")

    Xs = make_features(src, w_pos)
    Xt = make_features(tgt, w_pos)

    Ns = Xs.shape[0]
    Nt = Xt.shape[0]
    ns_eff = min(ns, Ns)
    nt_eff = min(nt, Nt)
    print(f"[0/4] Sampling pixels for OT: ns={ns_eff}/{Ns},
    nt={nt_eff}/{Nt} (seed={seed})")

    idx_s = rng.choice(Ns, size=ns_eff, replace=False)
    idx_t = rng.choice(Nt, size=nt_eff, replace=False)
```

```
Xs_s = Xs[idx_s]
Xt_t = Xt[idx_t]
a = np.full(ns_eff, 1.0 / ns_eff, dtype=np.float64)
b = np.full(nt_eff, 1.0 / nt_eff, dtype=np.float64)

C = cost(Xs_s, Xt_t, cost_chunk)
P, err, iters = sinkhorn_log(a, b, C, eps, sinkhorn_iters,
sinkhorn_tol, check_every)

print(f"[3/4] Barycentric projection to colours
(Sinkhorn iters={iters}, final err={err:.3e})")

tgt_rgb = tgt.reshape(-1, 3).astype(np.float64)
Y = tgt_rgb[idx_t]
row_sums = np.maximum(P.sum(axis=1, keepdims=True), 1e-12)
barycentric_rgb_samples = (P @ Y) / row_sums

nn = nearest_neighbor_map(Xs, Xs_s, nn_chunk)
out_rgb = barycentric_rgb_samples[nn].reshape(Hs, Ws, 3)
.clip(0, 255).astype(np.float32)

save_rgb_255(out_path, out_rgb)
print(f"Done. Saved: {out_path}")
display_triptych(src, tgt, out_rgb,
titles=("Source", "Target", "Output"))
```

The above runs the colour transfer process. We set a random seed; import, load, and process source and target images; sample pixels; find the cost matrix; perform the Sinkhorn algorithm on the samples; Barycentrically project each sample pixel's output; map remaining pixels to their nearest neighbor and assign their new colour; then save and display the resulting image.

```
def load_rgb_255(path: str) -> np.ndarray:
    """Load image as float32 RGB in [0,255], shape (H,W,3)"""
    im = Image.open(path).convert("RGB")
    return np.asarray(im, dtype=np.float32)
```

This turns images into arrays.

```
def make_features(img_255: np.ndarray, w_pos: float)
    -> np.ndarray:
    """
    Per-pixel feature vectors:
    [R,G,B, w_pos*px, w_pos*py] where px,py in [0,255]
    (global coords)
    Returns float64 array (N,d).
    """
    H, W, _ = img_255.shape
    rgb = img_255.reshape(-1, 3).astype(np.float64)

    ys = np.repeat(np.arange(H), W).astype(np.float64)
    xs = np.tile(np.arange(W), H).astype(np.float64)

    px = np.zeros_like(xs) if W == 1 else 255.0 * xs / (W - 1)
    py = np.zeros_like(ys) if H == 1 else 255.0 * ys / (H - 1)
```

```
    pos = np.stack([px, py], axis=1)
    return np.concatenate([rgb, w_pos * pos], axis=1)
```

This add positions to image arrays whose entries are RBG values.

```
def cost(A: np.ndarray, B: np.ndarray, chunk: int)
    -> np.ndarray:
    """
    Finds cost (squared euclidean) between A and B.
    Uses chunking over A to control memory.
    Returns C shape (n,m), float64.
    """
    n, d = A.shape
    m, _ = B.shape
    C = np.empty((n, m), dtype=np.float64)
    B2 = np.sum(B * B, axis=1, keepdims=True).T
    print(f"[1/4] Building cost matrix C (shape {n}x{m})")

    for start in range(0, n, chunk):
        end = min(start + chunk, n)
        A_chunk = A[start:end]
        A2 = np.sum(A_chunk * A_chunk, axis=1, keepdims=True)
        #||a-b||^2=||a||^2+||b||^2-2a^Tb
        C_chunk = A2 + B2 - 2.0 * (A_chunk @ B.T)
        C[start:end] = np.maximum(C_chunk, 0.0)
```

```python
    return C
```

This finds the cost between two matrices using the Euclidean squared metric.

```python
def sinkhorn_log(a: np.ndarray, b: np.ndarray, C: np.ndarray,
                 eps: float, max_iters: int, tol: float,
                 check_every: int) ->
                 tuple[np.ndarray, float, int]:
    """
    Entropic OT via log-domain Sinkhorn.
    Returns (P, err, iters).
    """
    a = np.asarray(a, dtype=np.float64)
    b = np.asarray(b, dtype=np.float64)
    C = np.asarray(C, dtype=np.float64)

    n, m = C.shape
    loga = np.log(a)
    logb = np.log(b)
    logK = -C / float(eps)
    logu = np.ones(n, dtype=np.float64)
    logv = np.ones(m, dtype=np.float64)

    err = np.inf
    it = 0

    for it in range(1, max_iters + 1):
```

```
        logKv = logsumexp(logK + logv[None, :], axis=1)
        logu = loga - logKv
        logKTu = logsumexp(logK + logu[:, None], axis=0)
        logv = logb - logKTu

        if it % check_every == 0 or it == max_iters:
            logKv = logsumexp(logK + logv[None, :], axis=1)
            logKTu = logsumexp(logK + logu[:, None], axis=0)

            r = np.exp(logu + logKv)
            c = np.exp(logv + logKTu)

            err_a = np.sum(np.abs(r - a))
            err_b = np.sum(np.abs(c - b))
            err = float(max(err_a, err_b))

            if err < tol:
                break

    P = np.exp(logu[:, None] + logK + logv[None, :])
    return P, err, it
```

This does Sinkhorn iterations until convergence on given source and target arrays.

```
def nearest_neighbor_map(A: np.ndarray, B: np.ndarray,
    chunk: int) -> np.ndarray:
    """
```

```
    For each pixel in A, finds nearest pixel in B
    by squared Euclidean distance.
    Returns idx shape (N,), int.
    Chunked over A with progress.
    """
    N, d = A.shape
    M, _ = B.shape
    idx = np.empty(N, dtype=np.int64)

    B2 = np.sum(B * B, axis=1)  # (M,)

    print(f"[4/4] Nearest-neighbor mapping: {N} points
    -> {M} samples")

    for start in range(0, N, chunk):
        end = min(start + chunk, N)
        A_chunk = A[start:end]
        A2 = np.sum(A_chunk * A_chunk, axis=1)
        D = A2[:, None] + B2[None, :] - 2.0 * (A_chunk @ B.T)
        idx[start:end] = np.argmin(D, axis=1)

    return idx
```

This returns a mapping of pixels in an array to the nearest one of a subset of pixels from that array.

```
def save_rgb_255(path: str, img_255: np.ndarray) -> None:
```

```
    """Save float image to disk as uint8"""
    img = np.clip(img_255, 0, 255).astype(np.uint8)
    Image.fromarray(img, mode="RGB").save(path)
```

This saves an output image.

```
def display_triptych(src_255: np.ndarray, tgt_255: np.ndarray,
                     out_255: np.ndarray,
                     titles=("Source", "Target", "Output"))
                     -> None:
    """Display source/target/output"""

    fig, axes = plt.subplots(1, 3, figsize=(14, 5))
    for ax, img, title in zip(axes,
    [src_255, tgt_255, out_255], titles):
        ax.imshow(np.clip(img, 0, 255).astype(np.uint8))
        ax.set_title(title)
        ax.axis("off")
    plt.tight_layout()
    plt.show()
    return
```

This displays the source, target, and new images to compare.

7.0.3 Colour Transfer Results

The following are all results from colour transfer. All images had $408 \times 612 = 249,696$ pixels, 6000 pixels were sampled from all source and targets, and all spatial weights were set to 0.5. All transports maxed out at 100 Sinkhorn iterations, meaning more iterations were necessary to reach our original desired level of convergence.

Three images were used as source and target images: green field (green field below a cloudy-blue sky), black & white field (a desaturated version of green field), and orange & purple (a random pattern of dark purple and bright orange).

Figure 7.1: Green field to orange & purple

This transfer was as expected - the dark parts of the green field image turned purple (the dark part of orange & purple), while the light parts turned orange (the light part of orange & purple). Darker and cooler colours have more similar RGB values with each other than they do with light and warm colours. The vice-versa is also true.

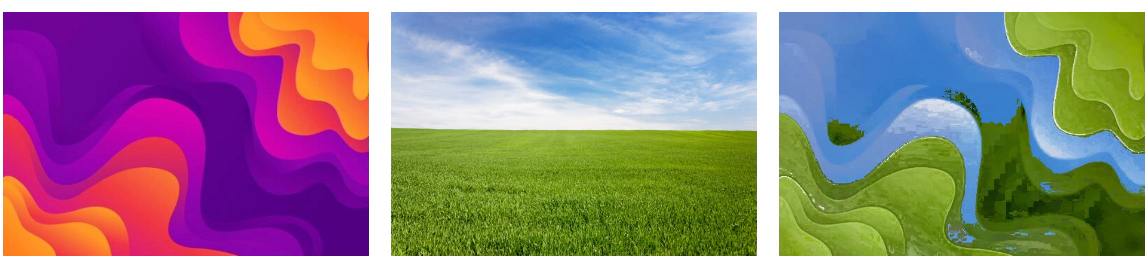

Figure 7.2: Orange & purple to green field

This transfer was also as expected, mapping light to dark areas.

Figure 7.3: Green field to black & white field

This transfer desaturated the green field, which ended up being more crisp than the target - meaning that lights got lighter and darks got darker.

Figure 7.4: Black & white field to green field

This transfer colourized the desaturated field. While mostly successful in colour matching, we see two notable spots of noise - where the sky is green and the field is blue. The mismatched spots occurred at the darkest area of blue sky and the lightest area of green field. It makes sense that this would happen, it would be cheaper to do this than transferring to their intended colours.

Chapter 8

Conclusion

The Optimal Transport Problem studies how to transform one probability distribution into another in the most cost-efficient way. Two formulations were explored. The Monge formulation uses transport maps, so each unit of mass is sent to a single target, which prevents mass splitting and can limit applicability. The Kantorovich formulation allows for transport plans, which distribute mass across multiple targets and guarantee the existence of solutions under broad conditions.

Several solution methods were examined. In the one-dimensional continuous case with a convex cost function, optimal transport admits a solution by matching quantiles via inverse cumulative distribution functions. For discrete problems with equally weighted points, the Monge formulation reduces to finding an optimal bijection, which can be identified through permutations of the source set. In the Kantorovich setting, Birkhoff's theorem implies that optimal transport plans correspond to permutation matrices, allowing the problem to be solved by searching over these structures.

Optimal transport can be formulated as a linear program. Classical methods, such as the simplex algorithm, can be used to compute exact solutions for discrete distributions, though continuous problems must first be discretized. While these approaches are precise, they can become computationally expensive for large-scale problems. To address this, regularization techniques introduce an entropy term, producing a smoother and more tractable problem. This leads to the Sinkhorn algorithm, which efficiently approximates optimal transport plans through iterative updates derived from Lagrangian conditions. Although approximate, this approach significantly improves computational performance.

The applicability of optimal transport was demonstrated through a colour transfer problem. By representing pixel colours as probability distributions, optimal transport provides a way to change an image's colour palette while preserving its structure. This example highlights how this theory can be translated into practical tools. While the implementation was successful, several limitations remain. The algorithm exhibited slow performance despite optimization attempts, suggesting that further improvements in numerical methods or implementation strategies could be beneficial. Additionally, some results contained noise or mismatched colours, indicating that more refined cost functions or regularization techniques may improve output quality.

This thesis was primarily guided by two foundational texts: Thorpe's *Introduction to Optimal Transport* [13], which provided the theoretical framework, and Peyré and Cuturi's *Computational Optimal Transport*, which introduced practical algorithms. Further exploration of topics such as the existence of transport maps and Wasserstein distances, as well as more advanced computational methods, would provide valuable extensions to this work. Overall, this thesis establishes a solid foundation in both the theory and implementation of optimal transport, while also identifying meaningful directions for further research and development.

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